

Technologies for E-Commerce

CAN302

**Department of Communications and Networking
Xi'an Jiaotong-Liverpool University (XJTLU)**

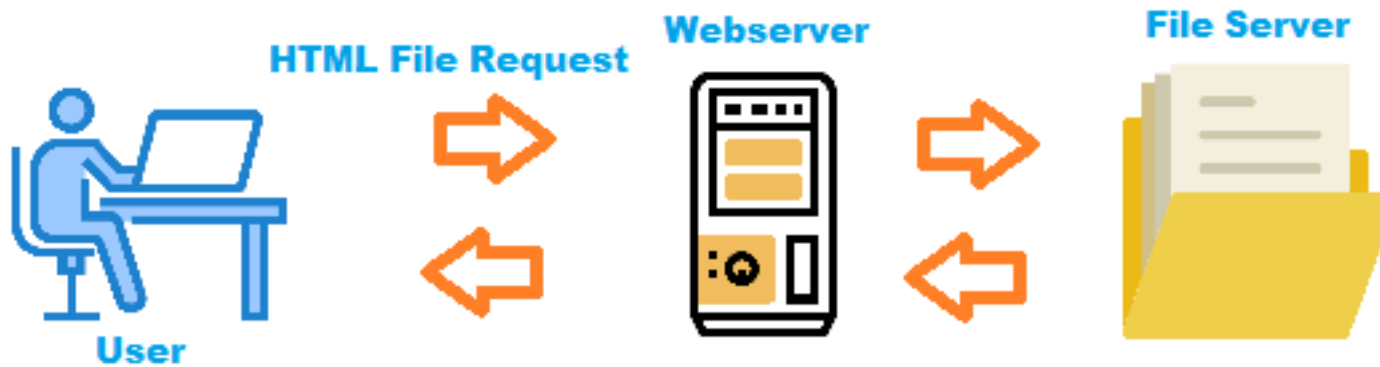
Lecture 4 – Insight into PHP & SQL



Outline: **@PHP** **@SQL**

WEB server – store the content remotely

Static Webpage Serve by Webserver



- A basic HTTP server is a software that understands URLs (web addresses) and send the corresponding file stored on the server computer to client through HTTP response

Each product for a HTML page ...

International Shopping: Shop E x Cloth Deals on eBay x +

amazon.com/b?node=16225009011&pf_rd_r=HV3Z0S5VDB26VZ87B413&pf_rd_p=5232c45b-5929-4ff0-8eae-5f67afd5c3dc&pd_rd_r=413e10b8-4e79-467d-92e5-013ed661db8a&pd_rd_w=BHgPy&pd_rd_w...

Department

- < Accessories & Supplies
- < Camera & Photo
- < Car & Vehicle Electronics
- < Cell Phones & Accessories
- < Computers & Accessories
- < GPS, Finders & Accessories
- < Home Audio
- < Office Electronics
- < Portable Audio & Video
- < Security & Surveillance
- < Television & Video
- < Video Game Consoles & Accessories
- < Video Projectors
- < Wearable Technology
- < eBook Readers & Accessories
- < Headphones, Earbuds & Accessories
- < Household Batteries, Chargers & Accessories
- < Power Accessories
- < Tablets

More Sustainable Products

- ☐ Climate Pledge Friendly

Payment Plan

- ☐ Layaway Eligible

Amazon Certified

- ☐ Auto Replenishment
- ☐ Works with Alexa

From Our Brands

- ☐ Amazon Brands

Featured Brands

- ☐ Amazon Basics
- ☐ Apple
- ☐ Allun
- ☐ HP
- ☐ Energizer
- ☐ Acoyer
- ☐ Syntech

Packaging Option

Electronics

Shop home entertainment, TVs, home audio, headphones, cameras, accessories and more

1-12 of over 70,000 results for **Electronics**

Amazon's Choice



Apple Pencil (2nd Generation)
4.8 ★★★★★ ~ (73,052)
\$89.99 \$129.00
More Buying Choices
\$78.96 (13 used & new offers)

Best Seller



Amazon Basics 48 Pack AA High-Performance Alkaline Batteries, 10-Year Shelf Life, Easy to Open Value Pack
48 Count (Pack of 1)
4.7 ★★★★★ ~ (677,349)
Amazon brand

Best Seller



Amazon Basics 36 Pack AAA High-Performance Alkaline Batteries, 10-Year Shelf Life, Easy to Open Value Pack
36 Count (Pack of 1)
4.7 ★★★★★ ~ (647,524)
Amazon brand



Apple EarPods Headphones with Lightning Connector. Microphone with Built-in Remote to Control Music, Phone Calls, and Volume. Wired Earbuds for...
4.6 ★★★★★ ~ (191,362)

Best Seller

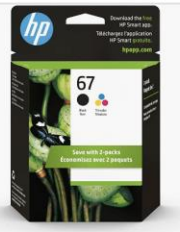


Apple 20W USB-C Power Adapter - iPhone Charger with Fast Charging Capability, Type C Wall Charger
4.8 ★★★★★ ~ (173,074)

Best Seller



Original HP 67XL Black High-yield Ink Cartridge | Works with HP DeskJet 1255, 2700, 4100 Series, HP ENVY 6000, 6400 Series | Eligible for Instant Ink |...
4.7 ★★★★★ ~ (33,388)



Original HP 67 Black/Tri-color Ink Cartridges (2 Count - Pack of 1) | Works with HP DeskJet 1255, 2700, 4100 Series, HP ENVY 6000, 6400 Series |...
4.7 ★★★★★ ~ (36,010)



Energizer AA Batteries, Double A Long-Lasting Alkaline Power Batteries, 32 Count (Pack of 1)
32 Count (Pack of 1)
4.8 ★★★★★ ~ (15,125)

Best Seller



Seagate Portable 2TB External Hard Drive HDD — USB 3.0 for PC, Mac, PlayStation, & Xbox -1-Year Rescue Service (STGX2000400)
4.7 ★★★★★ ~ (197,670)



Beats Solo3 Wireless On-Ear Headphones - Apple W1 Headphone Chip, Class 1 Bluetooth, 40 Hours of Listening Time, Built-in Microphone...
4.7 ★★★★★ ~ (61,541)



Roku Streaming Stick 4K | Streaming Device 4K/HDR/Dolby Vision with Roku Voice Remote and TV Controls
4.7 ★★★★★ ~ (50,112)
\$39.99 \$49.99

Best Seller



Amazon Basics 8 Units Pack 9 Volt Performance All-Purpose Alkaline Batteries, 5-Year Shelf Life, Easy to...
8 Count (Pack of 1)
4.7 ★★★★★ ~ (143,090)

- Manually updating the html pages would be a nightmare!

Dynamic webpage: frame + data



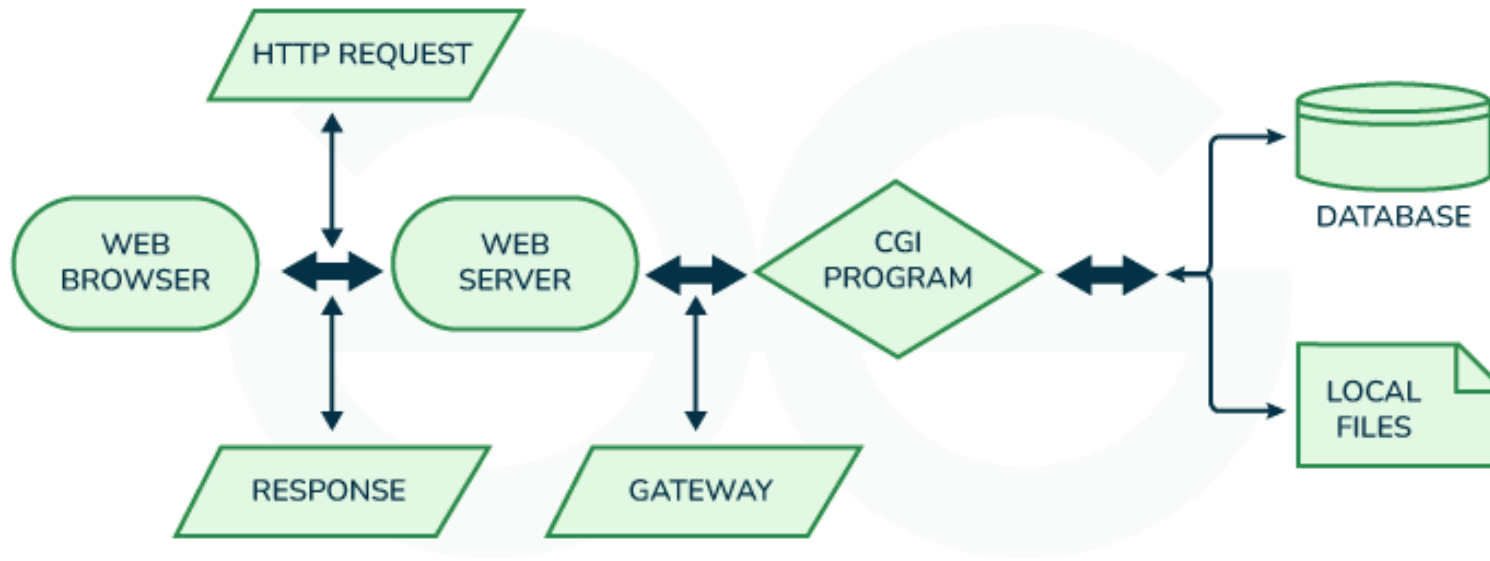
- Like a supermarket, we need to put products on the shelf in e-market.
- Manually update the products is not a possible way.
- It's time for dynamic web:

A dynamic website is the webpage on the server side where different contents are shown when assessed at different timings.

<https://www.educba.com/dynamic-website/>

<https://blog.csdn.net/dhp1027/article/details/7725653>

CGI - Common Gateway Interface



- CGI is an interface that enables the web server to process a user's HTTPS request by executing a program.
- CGI is only a standard (like HTTP) but not a program implementation.
- Server language can be C/C++, Perl, JAVA ..., the CGI program resides on the server to process data to produce the relevant dynamic content.
- CGI program can access the data stored in database

<https://www.geeksforgeeks.org/common-gateway-interface-cgi/>

Most of them can be applied for web server



So, how to choose the programming language?

Sever page



- ASP (Active Sever Page), 1996 by Microsoft.
 - Using VBScript/JavaScript to generate the dynamic html content
 - Calling the com components to extend its ability.
 - ASP.NET supports .NET languages, such as: C#, J#
- JSP (Java Server Page), 1998 by SUN.
 - It is JAVA but **NOT** JavaScript.
 - Html + JAVA code in one jsp file.
- PHP
 - 1.0 in 1994 by Rasmus Lerdorf, as Personal Home Page Tools, a **CGI** with Perl like and shell language.

Glance at ASP

```
1. <html>
2. <body>
3. <%
4. response.write("Hello World!")
5. %>
6. </body>
7. </html>
```

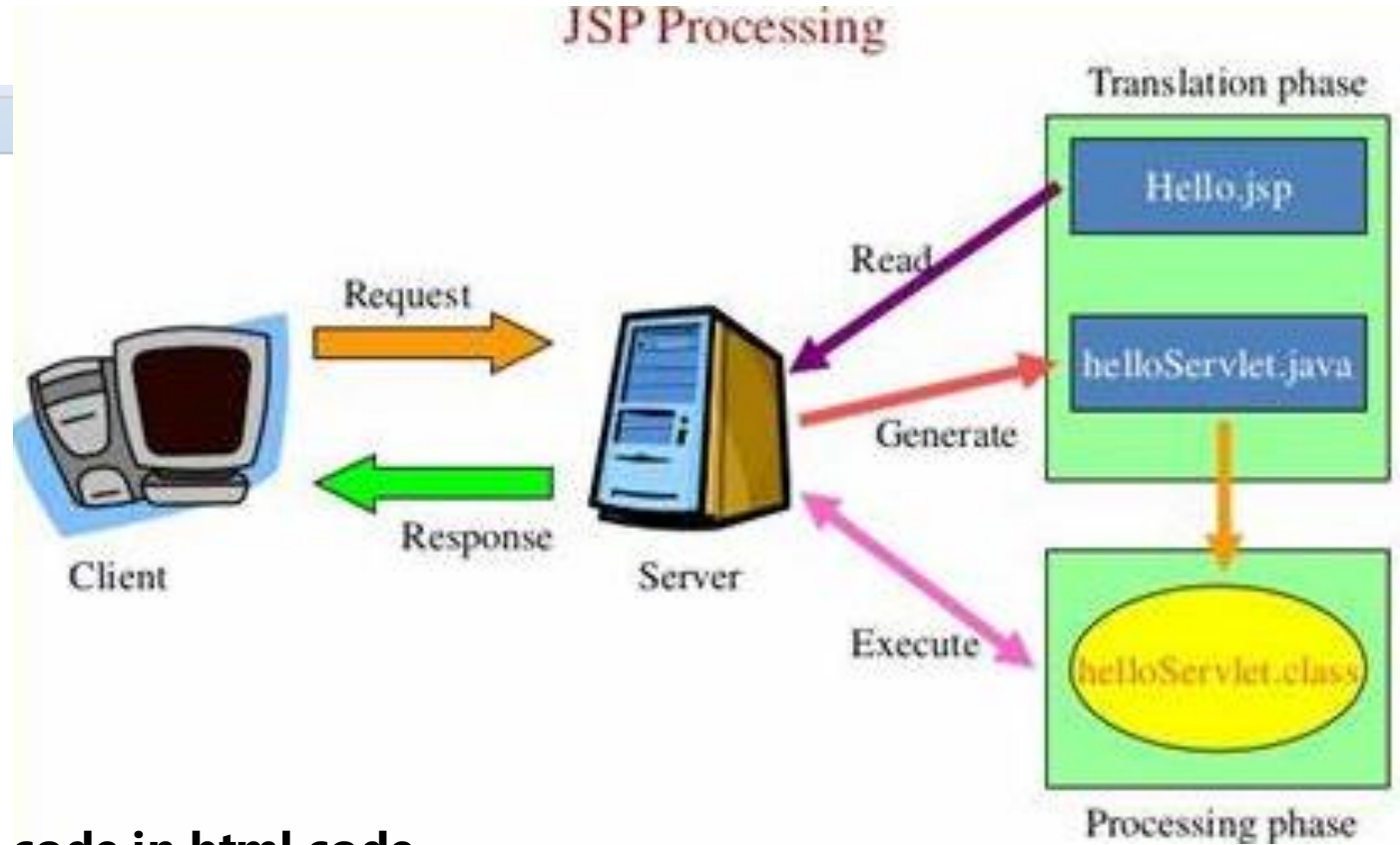
- Embedded VB or JS code in html file.
- The dynamic info (such as products, customers, orders...) can be stored in database.
- On request, the **web server** will run Vb or JS code (such as trace info from database) and generate the corresponding real-time html codes.
- Then, web server will merge the static codes and dynamic generated codes as a whole html and sent it back to the browser.

- **Benefit:** **No need so much print...**
- **Drawback:** **The backend code and front-end code are mixing together**

<https://zhuanlan.zhihu.com/p/42343690>

JSP, more programmer friendly than JAVA

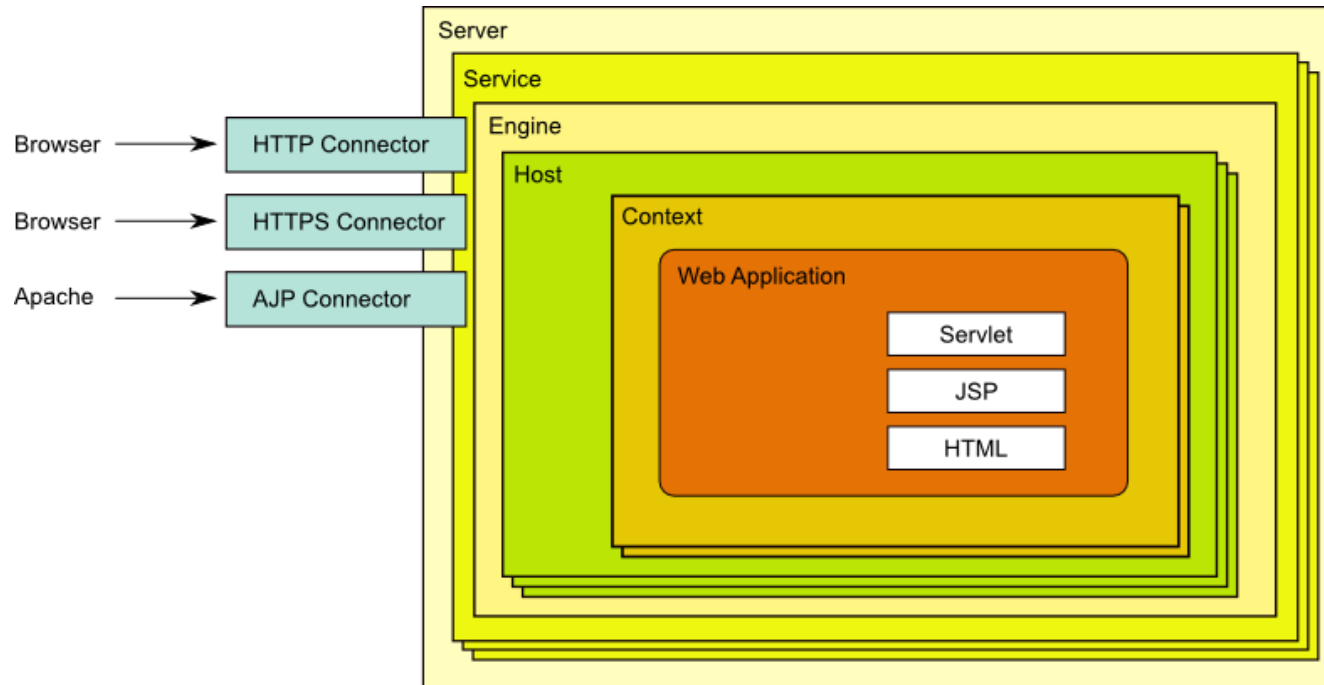
```
Scripting.jsp  Scripting Element Demo
1  <%@ page language="java" contentType="text/html;
2  charset=ISO-8859-1" pageEncoding="ISO-8859-1"%>
3
4  <!DOCTYPE html>
5  <html>
6  <head>
7  <meta charset="ISO-8859-1">
8  <title> Scripting Element Demo </title>
9  </head>
10 <body>
11 <%! int a = 10; %>
12 <% out.println(" The value of a is: " +a ); %>
13 </body>
14 </html>
```



- **Embed the java code in html code.**
- **Auto transfer it to Java code for compiling.**

<https://blog.csdn.net/jie1521836/article/details/81323942>
<https://zhuanlan.zhihu.com/p/42343690>

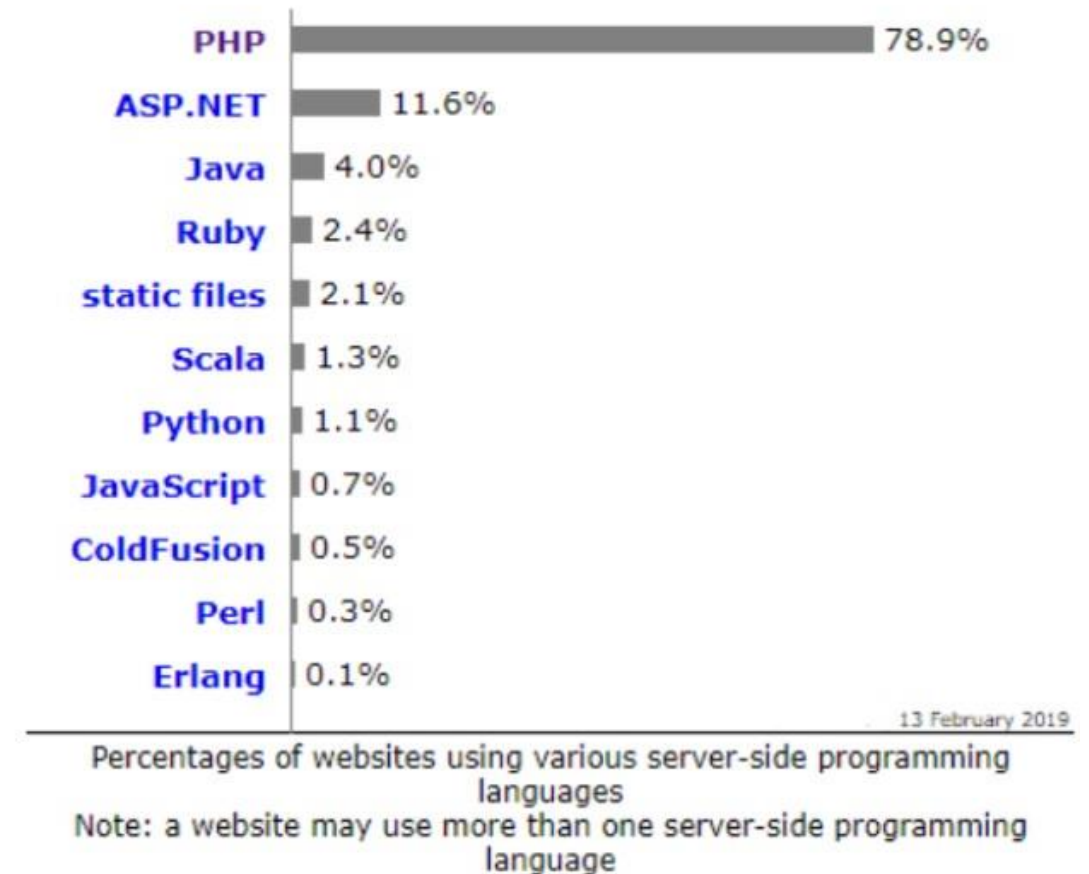
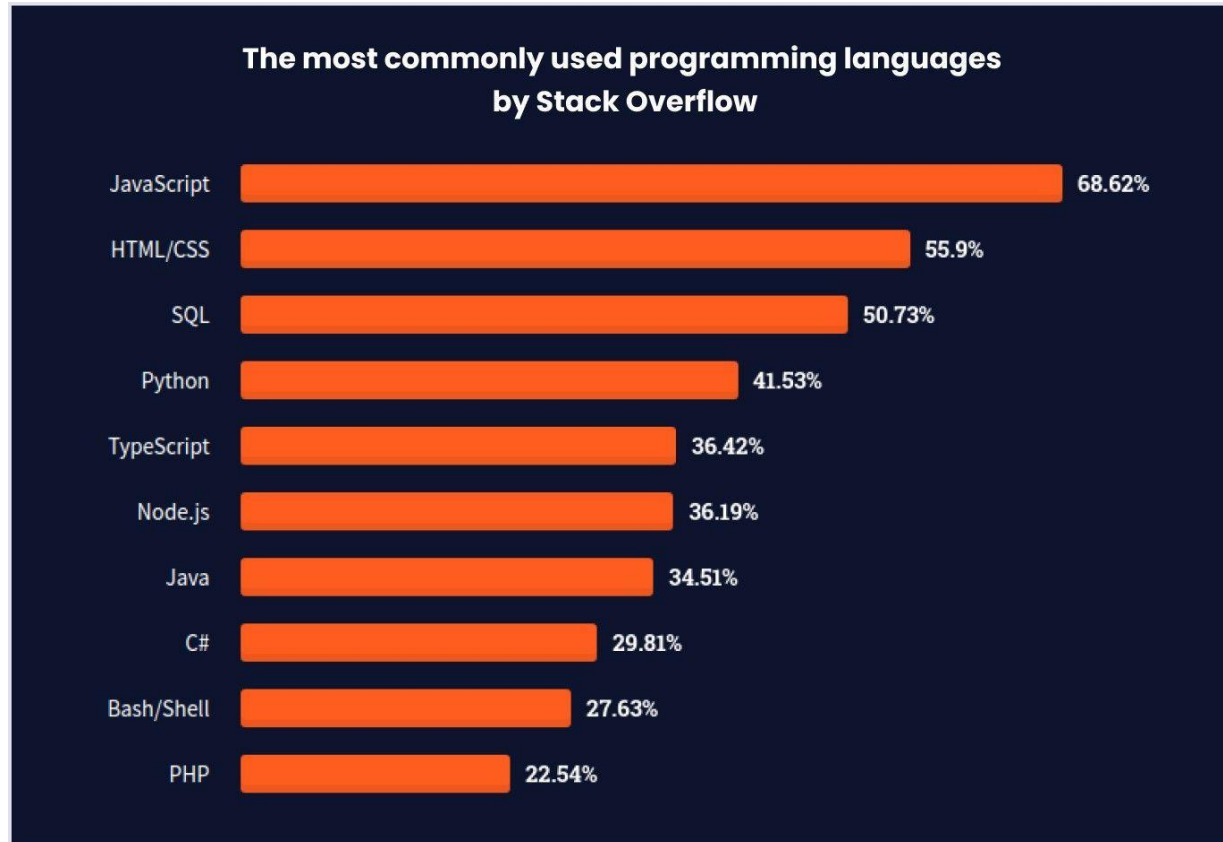
Tomcat – Web server for JAVA



- The Apache Tomcat **server** is Java-based, open source reference implementation (RI) web application and servlet container created to run servlets and Java Server Pages (JSP) based web applications.
- Again, Tomcat server is a **software**.
- It can work together with Apache server.

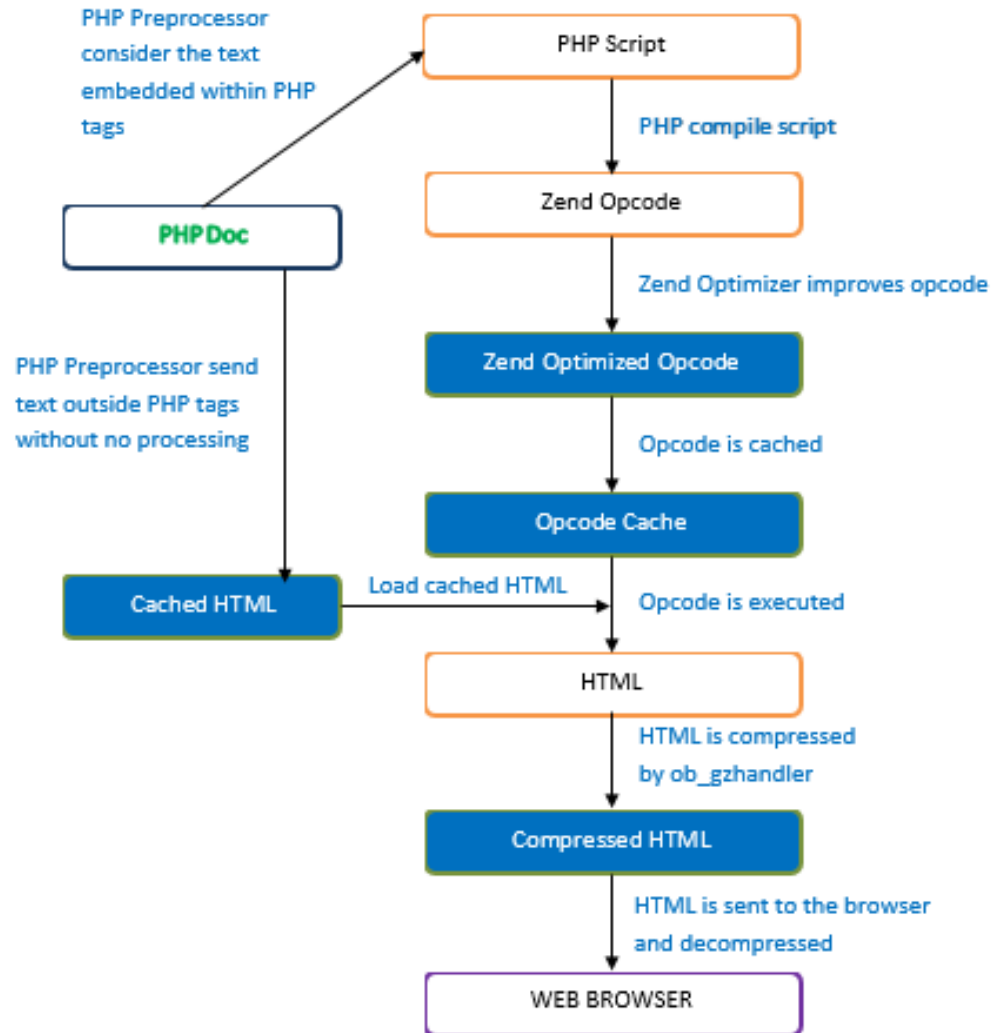
<https://www.jvmhost.com/tomcat-hosting/>

Confusing data



- **Why PHP is not a popular language but widely used by so many websites?**

PHP - Hypertext Preprocessor



In 1997, Andi Gutmans and Zeev Suraski rewrote Rasmus Lerdorf's PHP-FI, which was released as PHP Hypertext Preprocessor 3.

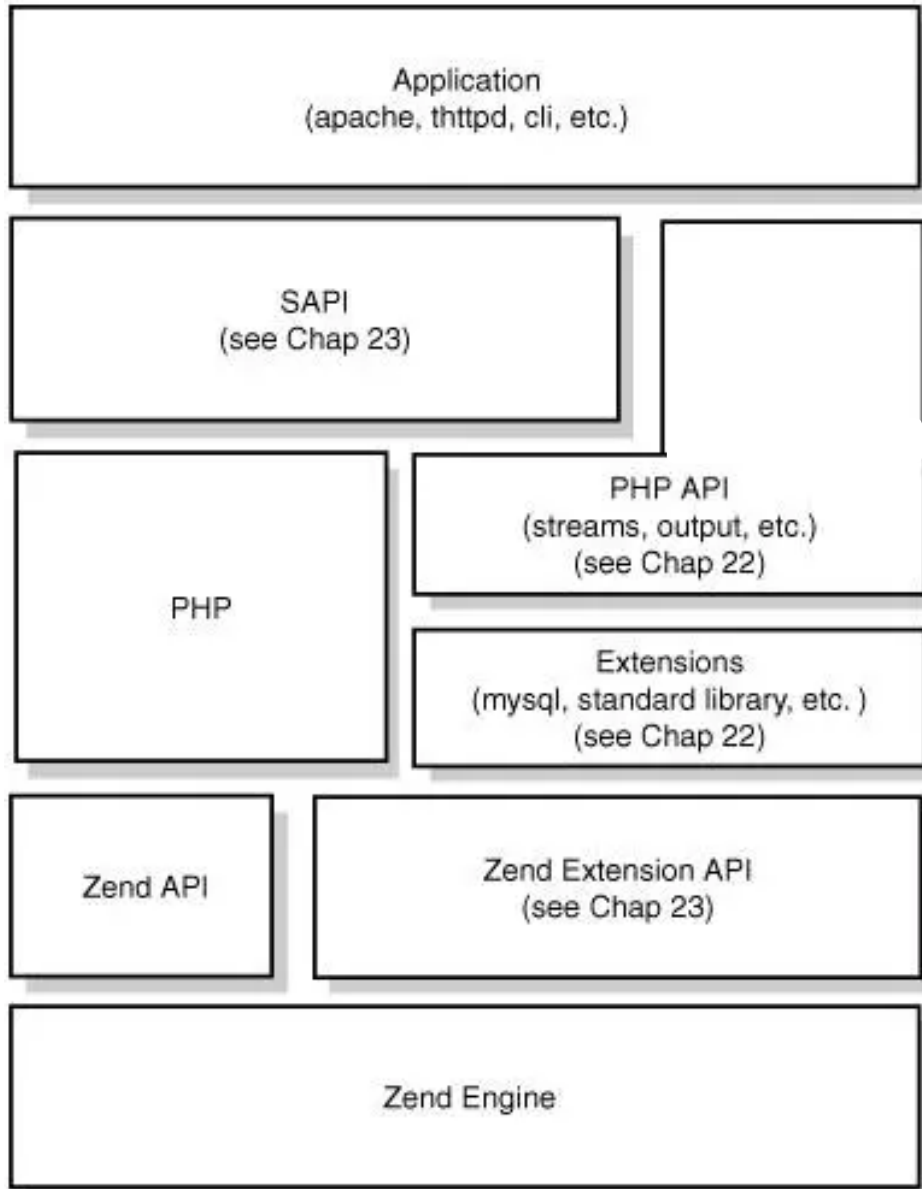
In 1998, they completely redesigned the PHP syntax parser and named it the Zend Engine. PHP 4 was the first product based on the Zend engine, which also achieved great success.

In 1999, they co-founded Zend Corporation.

The PHP scripts are embedded in html codes with opening tag `<?php` and closing tag `?>`

When a PHP page is requested by the client, Web server (like Apache) will interpret the PHP scripts and use the output to replace the PHP scripts, the dynamic generated HTML content will be returned to the client.

PHP – architecture

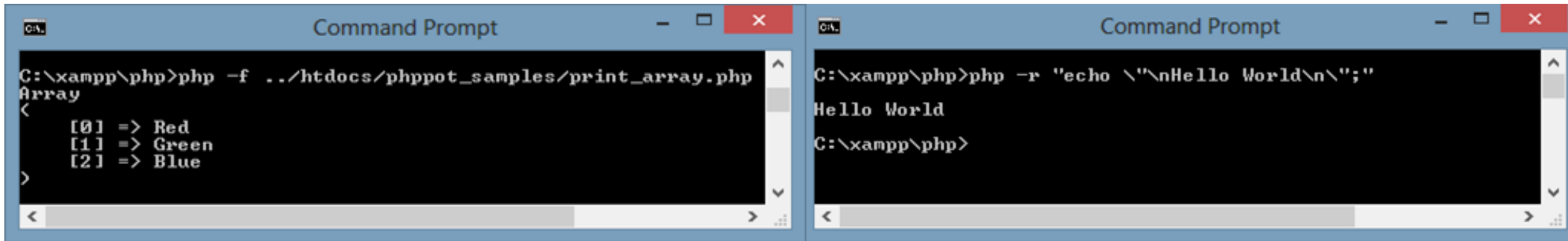


- PHP supports to embedded the scripts in html codes.
- PHP is optimized for Web development and can be used for general-purpose (However few people use it beyond web).
- PHP is the widely-used, free, and still active.
- Nginx, Apache have extension that support PHP through SAPI of PHP.

SAPI of PHP

- [CLI - command line interface](#)
 - Using PHP in command line interface.
- [phpdbg - interactive PHP debugger](#)
 - PHP includes an interactive debugger called phpdbg implemented as a SAPI module.
- [CGI - common gateway interface](#)
- [Embed](#)
- [FPM](#)
- [Litespeed](#)
- [Apache2Handler](#)
- **Server Application Programming Interface** - SAPI helps interacting between the outside world and the PHP/Zend engine.
- PHP comes with multiple SAPI modules for using it in command line, with web servers or even embedding it in other applications.

PHP CLI



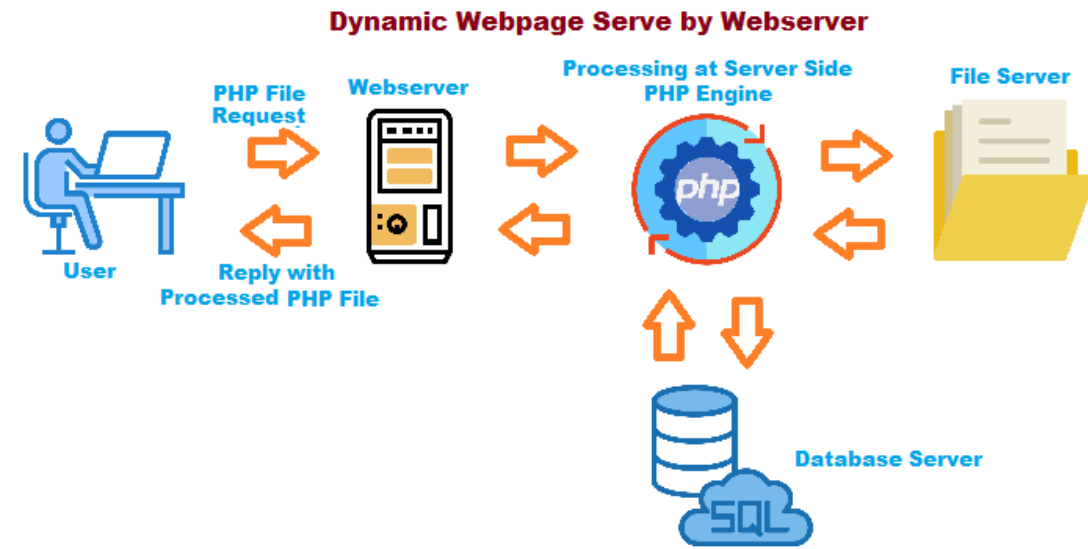
The image shows two side-by-side Windows Command Prompt windows. The left window displays the execution of a PHP script that prints an array. The right window displays the execution of a PHP command that echoes a formatted string.

```
C:\xampp\php>php -f ../htdocs/phppot_samples/print_array.php
Array
<
    [0] => Red
    [1] => Green
    [2] => Blue
>
```

```
C:\xampp\php>php -r "echo \"\nHello World\n\";"
Hello World
C:\xampp\php>
```

- PHP can run the code in command line, similar to most other program language.

PHP CGI for dynamic web



- Processed results are still H5+CSS3+JS
- While the same PHP file will output **DYNAMIC** result

Environment Variables Used in CGI

CONTENT_LENGTH : It specifies the length of URL encoded data in bytes.

REMOTE_HOST : resolved host name of the client.

CONTENT_TYPE : It specifies the type of data as a MIME header.

QUERY_STRING : Gives information at the end of the URL after ?

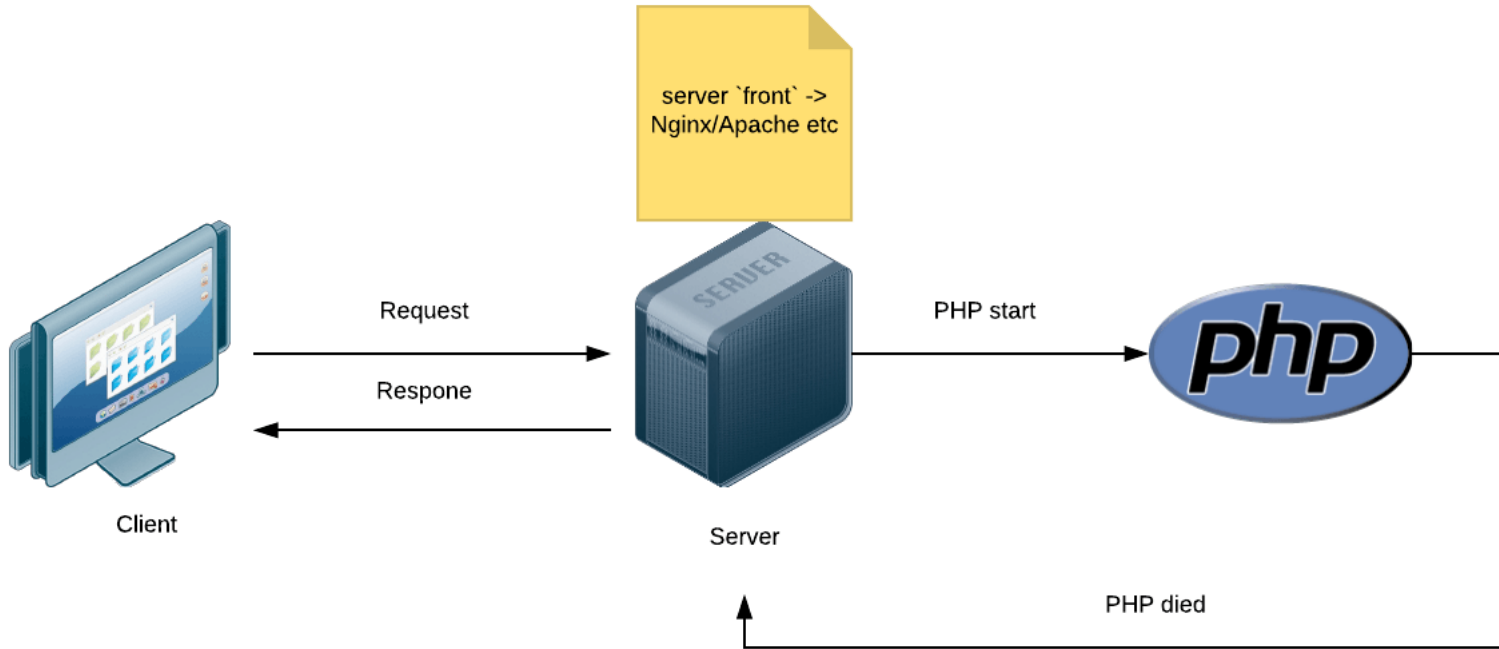
REMOTE_ADDR : IP address of the client making the request.

REQUEST_METHOD: It gives the HTTP request method.

SCRIPT_NAME : Name of the CGI script (i.e. PHP file path/name) which is being run.

<https://www.geeksforgeeks.org/what-is-the-role-of-cgi-in-web-development/>

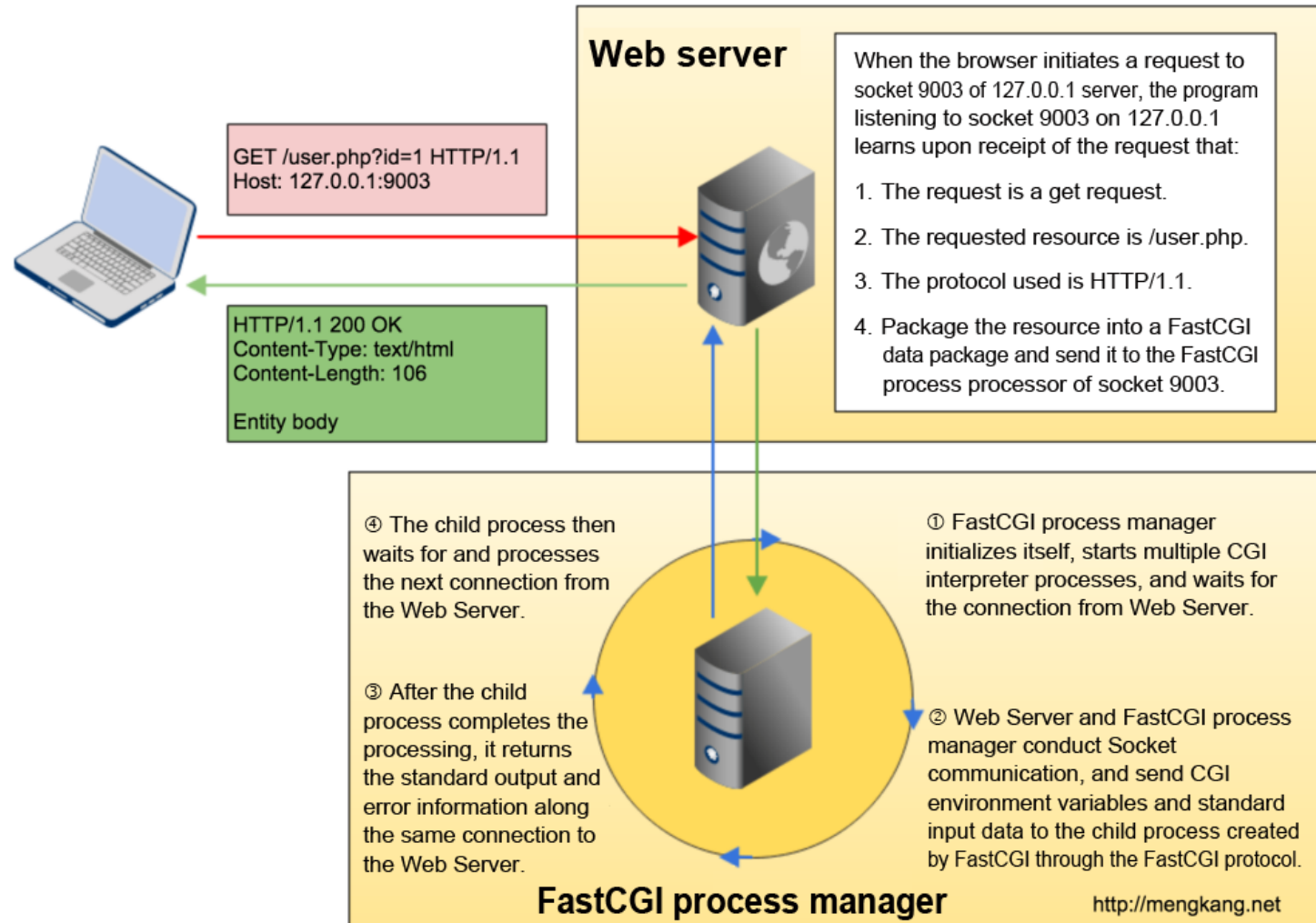
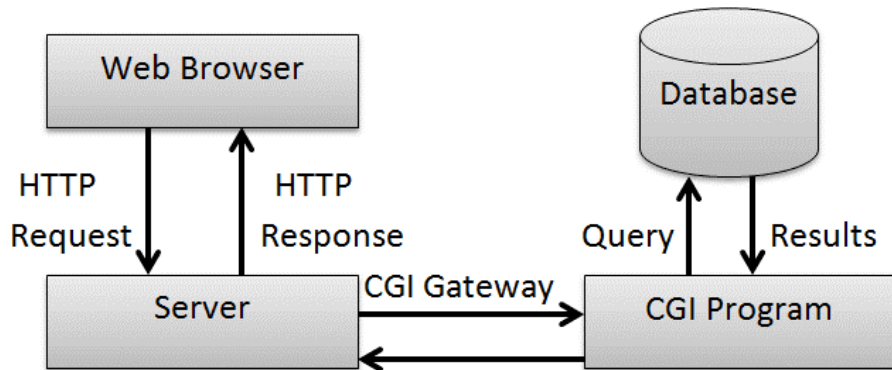
PHP - CGI lifecycle



- CGI is the SAPI of PHP to support the CGI standard.
- For each request, PHP does its job, then simply dies.

<https://tsh.io/blog/php-pm-guide-getting-started-with-the-process-manager/>

CGI and FastCGI

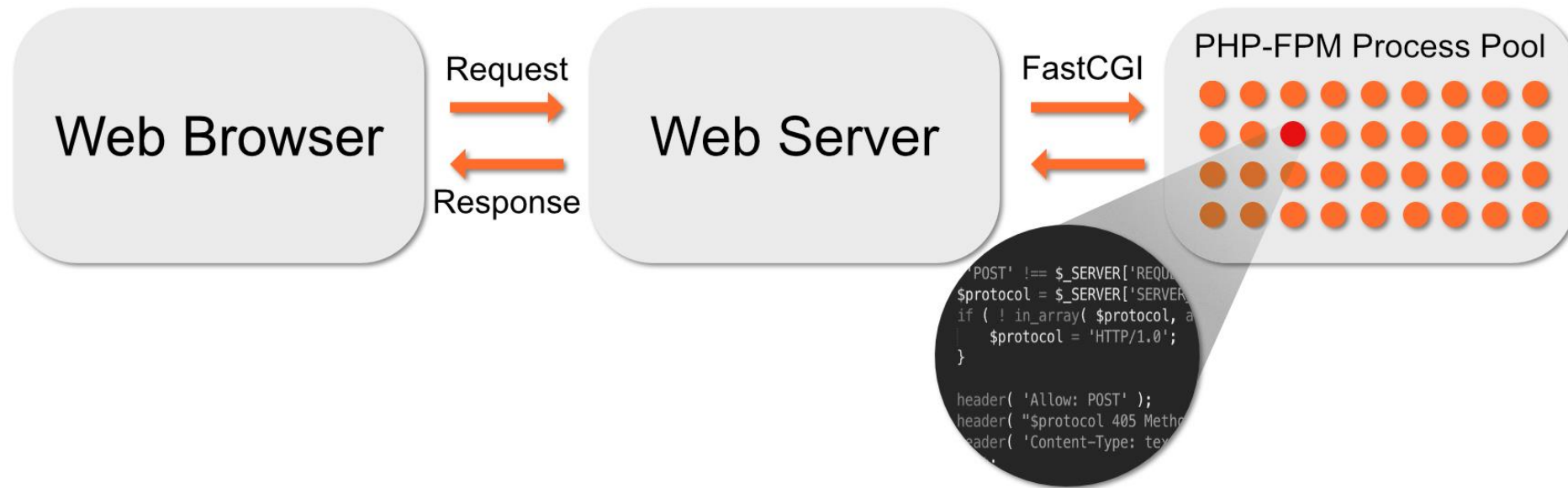


- FastCGI is an open extension to CGI that improve the performance.
- It saves the initial cost for each visiting.

<https://fast-cgi.github.io/original/>

https://www.semicolonandsons.com/code_diary/web-development/what-are-CGI-and-fast-CGI

PHP-FPM

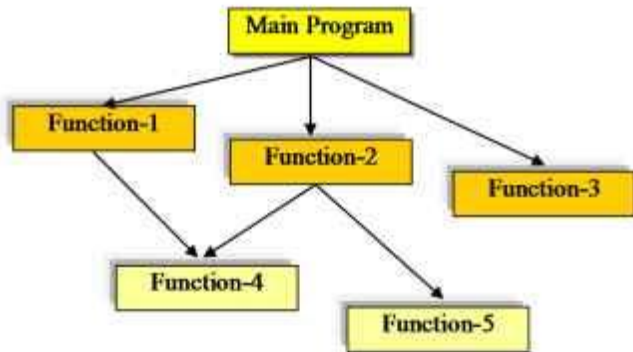


- A supervisor process chooses a worker process from the pool and gives it the code.
- The worker executes the code, and the result is sent back to Apache, which sends it to the web browser.
- Once the worker is done, it returns to the pool to await another chunk of code to execute.
- The pool size can be dynamically adjusted.

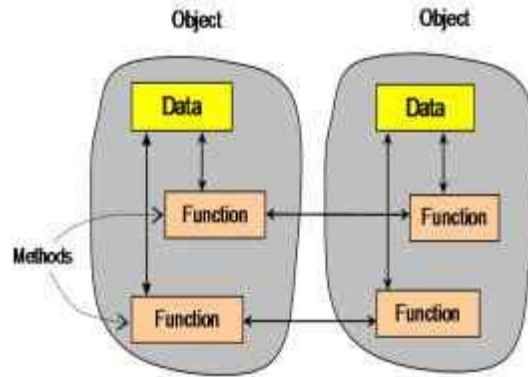
<https://cpanel.net/blog/tips-and-tricks/how-to-use-php-fpm-with-cpanel/>

PHP paradigm

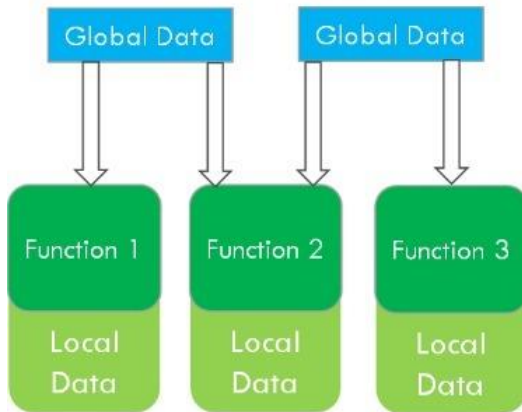
Procedure-oriented Programming



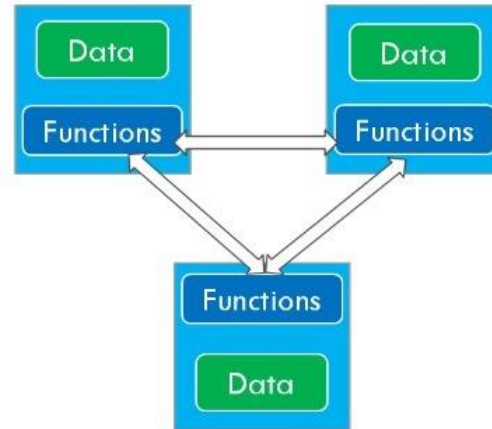
Object-oriented Programming



Procedural Oriented Programming



Object Oriented Programming



```

1  <?php
2  interface iABC {
3      public function __construct($name);
4      public function getName();
5  }
6  abstract class DEF {
7      public $email;
8      public function getEmail() {
9          return $this->email;
10     }
11     abstract protected function setEmail($email);
12 }
13 class XYZ extends DEF implements iABC {
14     protected $name;
15     public function __construct ($name) {
16         $this->name = $name;
17     }
18     public function setEmail($email) {
19         $this->email = $email;
20     }
21     public function getName() {
22         return $this->name;
23     }
24 }
    
```

PHP

abs

inte

- PHP supports both paradigms

<https://brainbell.com/php/abstract-classes-vs-interfaces.html>

PHP basic syntax

```
<?php
// PHP code goes here
?>
```

```
<!DOCTYPE html>
<html>
<body>

<h1>My first PHP page</h1>

<?php
echo "Hello World!";
?>

</body>
</html>
```

- PHP script can be placed anywhere in the document.
- A PHP script starts with <?php and ends with ?>
- "echo" is a **language construct** to output data.

Language construct



- In programming, language constructs and built-in functions are often misinterpreted with one another due to the fact that both have more or less alike behavior.
- But they differ from each other in the way the PHP interpreter interprets them.
- Every programming language consists of tokens and structures which the respective language parser can recognize.
- So whenever a file is parsed, the parser understands their usage and knows well what to do with them without having the need to examine them further.
- These tokens and structures are known as **language construct**. They are basically keywords that are a part of the programming language.
- In other words, they form the syntax of the language.

PHP key words

abstract	and	Array()	as	break
callable	case	catch	class	clone
cost	continue	declare	dafault	die()
do	echo	else	elseif	empty
enddeclare	endfor	endforeach	endif	endswitch
endwhile	eval()	exit()	extends	final
finally	for	foreach	function	global
goto	if	implements	include	include_once
instanceof	insteadof	interface	isset()	list()
namespace()	new	or	print	private
protected	public	require	require_once	return
static	switch	throw	trait	try
unset()	use	var	while	xor
yield	yield_from			

- Most of keywords function like its name.
- Similar to many other computer language.
- Not possible to explain all of them in the lecture.

About echo

```
$txt1 = "Learn PHP";  
$txt2 = "W3Schools.com";  
  
echo '<h2>' . $txt1 . '</h2>';  
echo '<p>Study PHP at ' . $txt2 . '</p>';
```

```
$txt1 = "Learn PHP";  
$txt2 = "W3Schools.com";  
  
echo "<h2>$txt1</h2>";  
echo "<p>Study PHP at $txt2</p>";
```

- All keywords are case non-sensitive including echo.
- Single quote and double quote are obviously different in PHP as above.
- You can use either one.

https://www.w3schools.com/php/php_echo_print.asp

PHP extensions

- PHP has basic internal functions or PHP core.
- Many functions are achieved through external components.
- Such as extension PDO (PHP Data Objects) let PHP can interact with most of the relational as well as NOSQL databases.

version: **7.2**

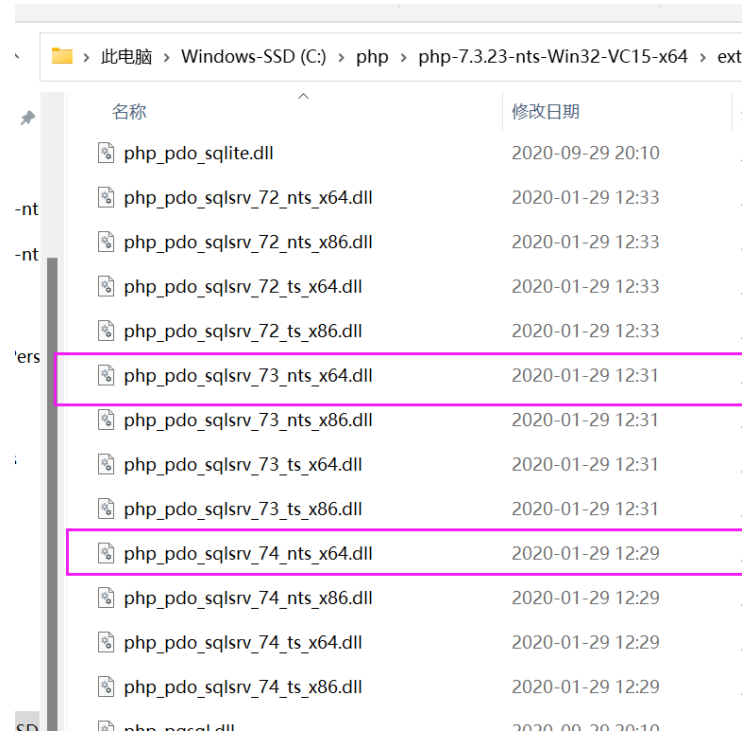
7.2 ▼ Set as current

[Switch To PHP Options](#)

☐ imagick	☐ newrelic	☒ posix	☐ tidy
☒ imap	☐ oauth	☐ propro	☐ timezonedb
☐ inotify	☐ oci8	☐ pspell	☐ trader
☐ interbase	☐ odbc	☐ raphf	☐ uploadprogress
☐ intl	☐ opcache	☐ redis	☐ uuid
☐ ioncube_loader	☐ pdf	☐ snmp	☐ vips
☒ json	☒ pdo	☐ soap	☒ wddx
☐ ldap	☐ pdo_dblib	☒ sockets	☐ xdebug
☐ lzf	☐ pdo_firebird	☐ sodium	☒ xmlreader
☐ mailparse	☒ pdo_mysql	☐ sourceguardian	☐ xmlrpc
☒ mbstring	☐ pdo_oci	☐ sqlsrv	☒ xmlwriter
☐ memcache	☐ pdo_odbc	☐ ssh2	☒ xsl
☐ memcached	☐ pdo_pgsql	☐ stats	☐ yaf
☐ mongodb	☒ pdo_sqlite	☐ swoole	☐ yaml
☒ mysqli	☐ pdo_sqlsrv	☐ sysvmsg	☒ zip
☒ mysqlind	☐ pgsql	☐ sysvsem	☐ zmq
☐ nd_mysql	☐ phalcon3	☐ sysvshm	
☐ nd_pdo_mysql	☒ phar	☐ tideways	

Defaults

Configure extensions



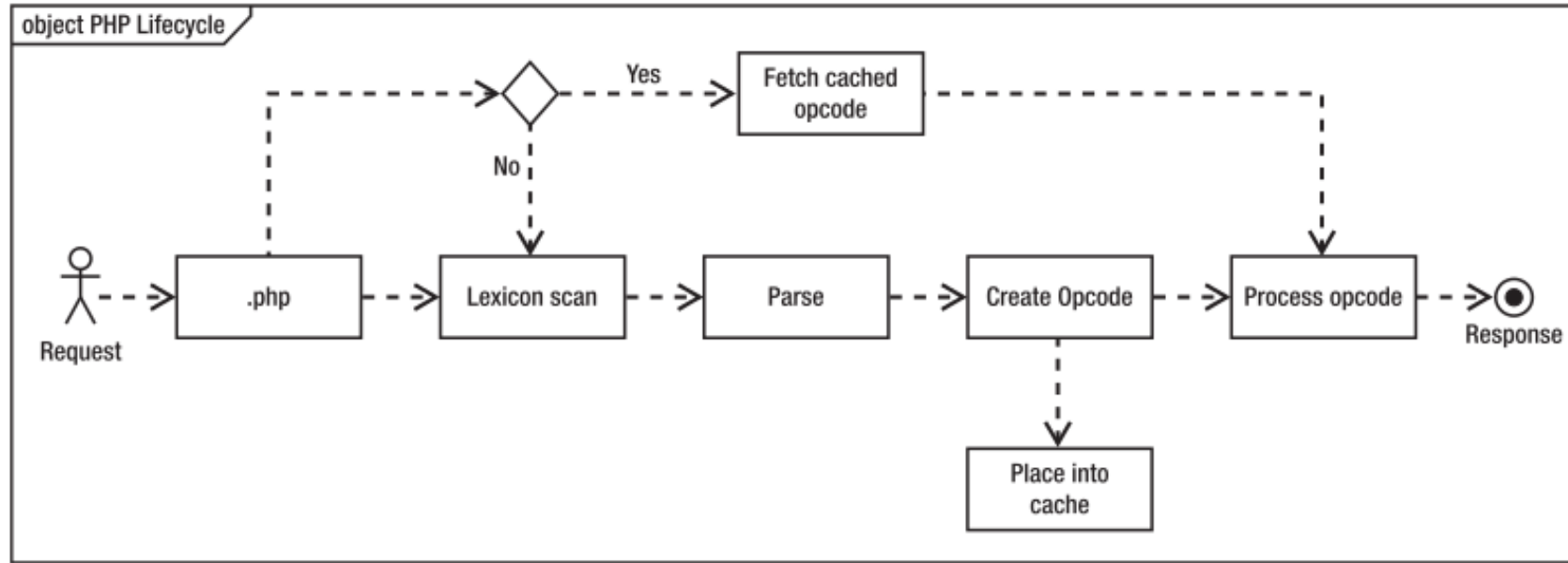
名称	修改日期
php_pdo_sqlite.dll	2020-09-29 20:10
php_pdo_sqlsrv_72_nts_x64.dll	2020-01-29 12:33
php_pdo_sqlsrv_72_nts_x86.dll	2020-01-29 12:33
php_pdo_sqlsrv_72_ts_x64.dll	2020-01-29 12:33
php_pdo_sqlsrv_72_ts_x86.dll	2020-01-29 12:33
php_pdo_sqlsrv_73_nts_x64.dll	2020-01-29 12:31
php_pdo_sqlsrv_73_nts_x86.dll	2020-01-29 12:31
php_pdo_sqlsrv_73_ts_x64.dll	2020-01-29 12:31
php_pdo_sqlsrv_73_ts_x86.dll	2020-01-29 12:31
php_pdo_sqlsrv_74_nts_x64.dll	2020-01-29 12:29
php_pdo_sqlsrv_74_nts_x86.dll	2020-01-29 12:29
php_pdo_sqlsrv_74_ts_x64.dll	2020-01-29 12:29
php_pdo_sqlsrv_74_ts_x86.dll	2020-01-29 12:29
php_pdo_sqlsrv_74_ts_x64.dll	2020-01-29 12:29
php_pdo_sqlsrv_74_ts_x86.dll	2020-01-29 12:29

```
929 ;extension=pdo_odbc
930 ;extension=pdo_pgsql
931 extension=pdo_sqlite
932 ;extension=pgsql
933 ;extension=shmop
934
935 ; The MIBS data available in the PHP distribution must be installed.
936 ; See http://www.php.net/manual/en/snmp.installation.php
937 ;extension=snmp
938
939 ;extension=soap
940 extension=sockets
941 ;extension=sodium
942 ;extension=sqlite3
943 ;extension=tidy
944 extension=xmlrpc
945 ;extension=xsl
946
947 extension=php_pdo_sqlsrv_73_nts_x64.dll
948 extension=php_sqlsrv_73_nts_x64.dll
949 client_buffer_max_kb_size = 50240
950 sqlsrv.ClientBufferMaxKbSize = 50240
951
952 ;;;;;;;;;;;;;;;;;;;;;;;;;
953 ; Module Settings ;
```

- Each PHP extension is a runnable program (.dll on Win and .so on Linux/Mac)
- PHP need to claim the specified extension in php.ini to load the corresponding extension.

<https://www.websitevidya.com/how-to-install-pdo-php-extension/>

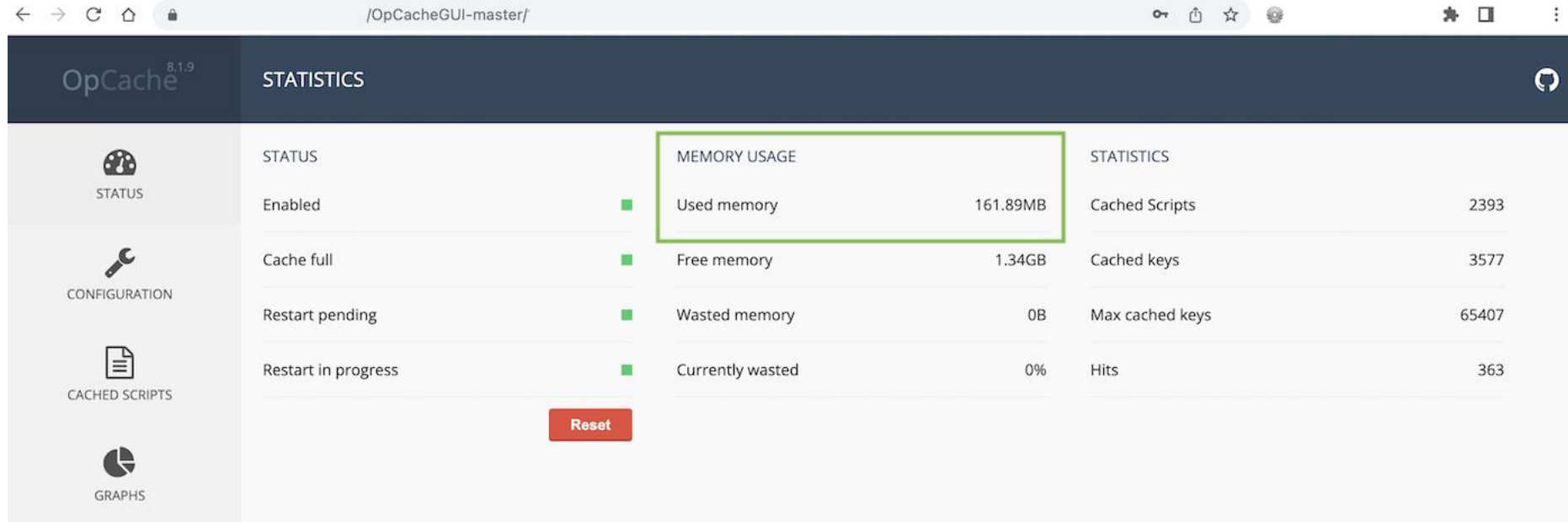
Zend Engine



- The Zend Engine is the open source scripting engine that interprets the PHP programming language.
- Zend Engine generates a series of operation codes, commonly known as "**opcodes**", representing the function of the code. Opcode is similar to bytecode in JS.
- It needs parse to abstract tree again before compile (similar to JS)!
- Opcodes can be executed by Zend executor.
- Cache is used to improve the efficiency.

<http://php.adamharvey.name/manual/it/internals2.opcodes.php>

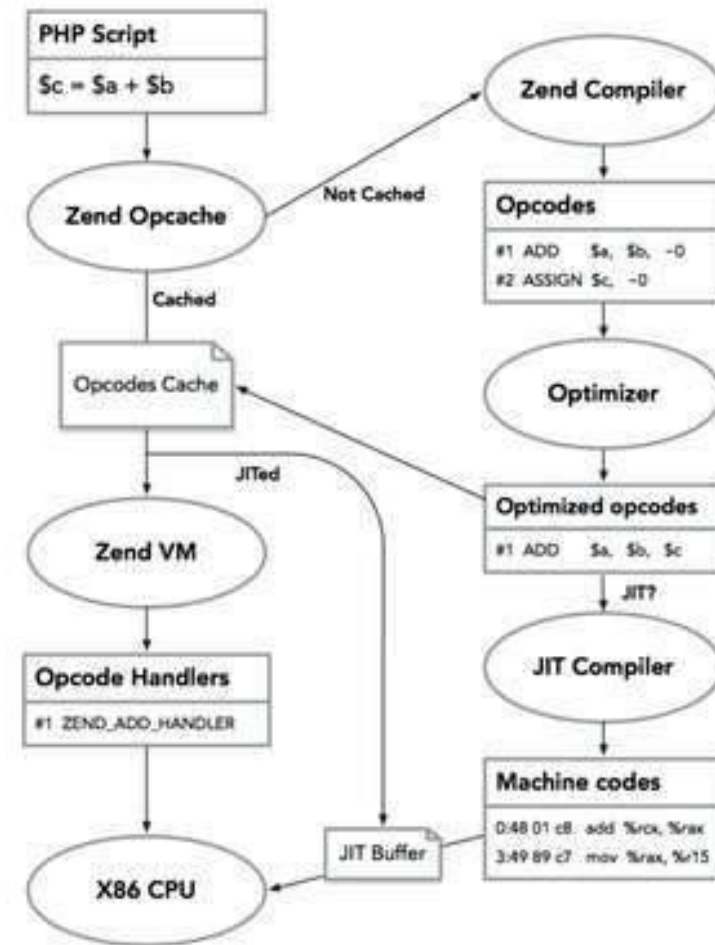
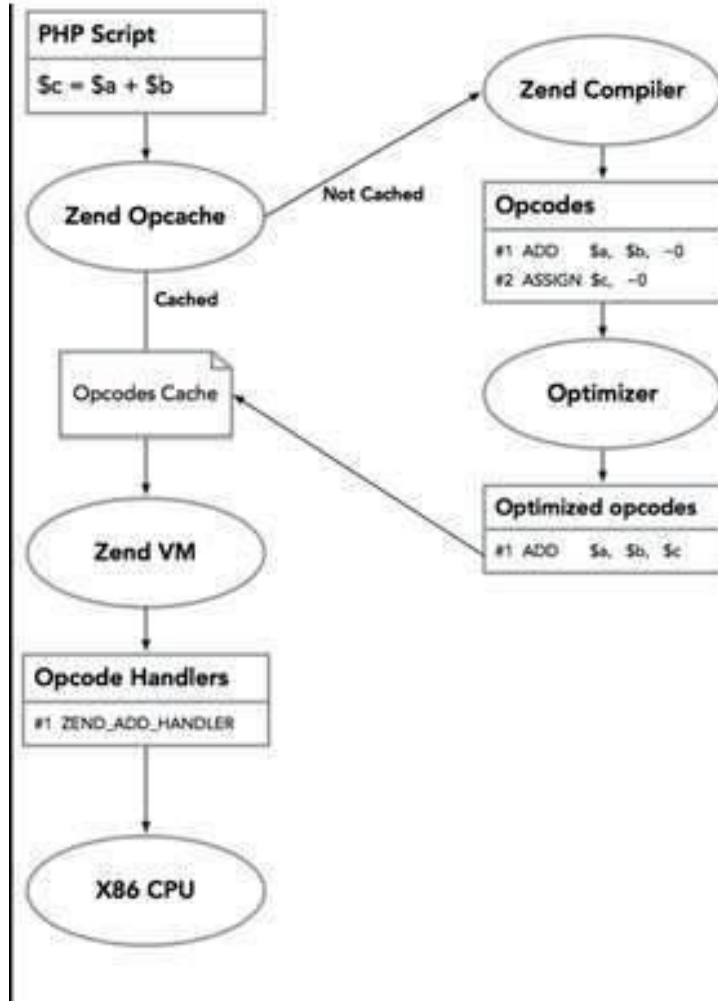
Zend Extension



- Zend Extensions are a set of libraries that provide additional functionality to the PHP programming language.
- They are designed to extend the core features of PHP and provide developers with advanced tools for building dynamic and high-performance web applications.
- By using Zend Extensions, developers can optimize their code, improve security, and enhance the overall user experience of their applications. i.e. OpcacheGUI can check cache status at graphic user interface.

<https://moldstud.com/articles/p-mind-blowing-zend-extensions-you-need-to-know-about>

Just in Time



- PHP also support **JIT** now!

Database basics



- Similar to **web server**, **database** is a software.
- A database is an organized collection of structured information, or data, typically stored electronically in a computer system.
- A database is usually controlled by a [database management system \(DBMS\)](#). Together, the data and the **DBMS**, along with the applications that are associated with them, are referred to as a database system.
- We often call a certain database which is actually the name of DBMS.

<https://www.oracle.com/database/what-is-database/#WhatIsDBMS>

Database vs Spreadsheet

Spreadsheets vs Databases

• Organize information in cells • Access to a limited amount of data • One user at a time • Control by user • Manual data entry 	• Organize information in complex collections • Access to a huge amount of data • Multiple users at a time • Control by DMS (Data Management System) • Strict & Consistent data entry 
---	--

relational databases





Spreadsheets

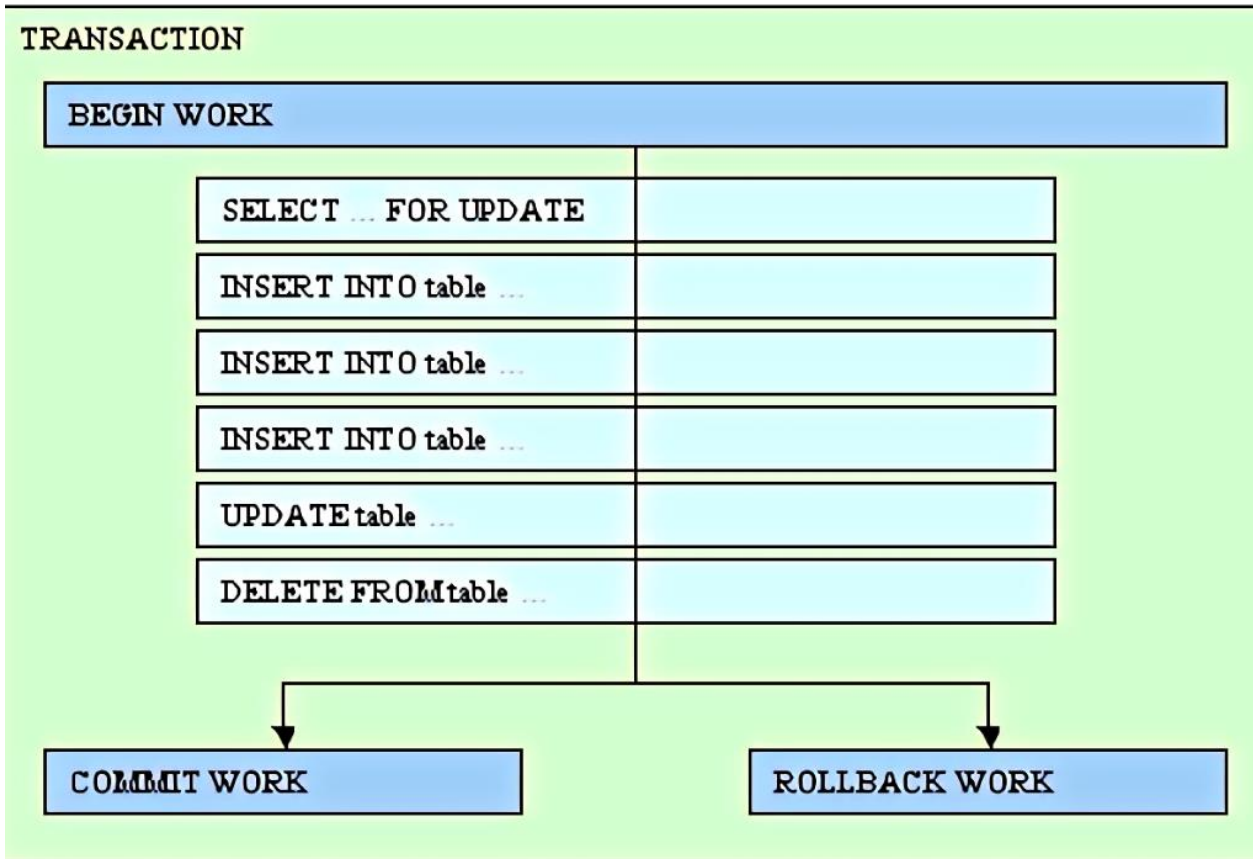
different cells can contain calculations
(functions and formulas)

all calculations and operations
are done after data retrieval

you can do calculations in "views"

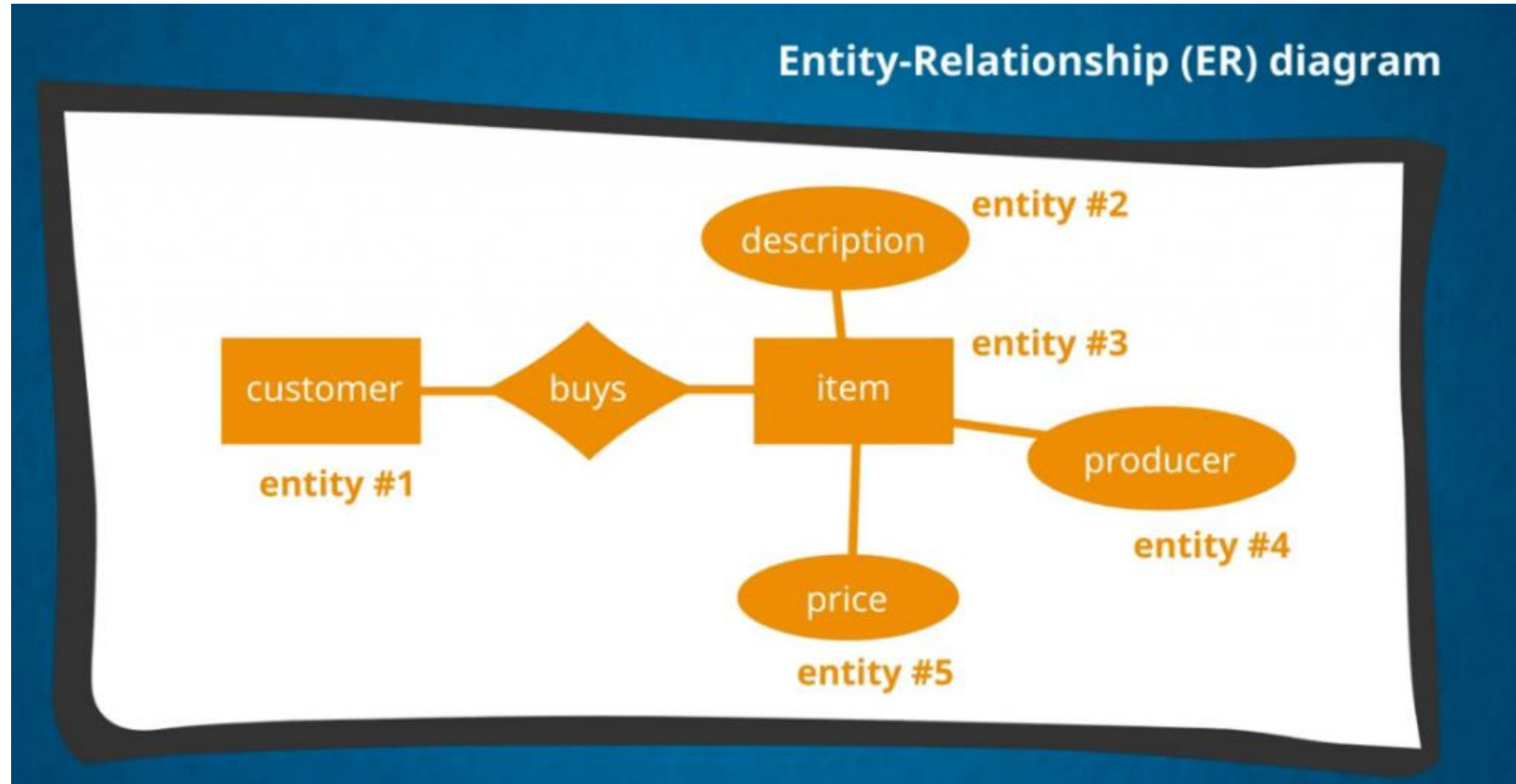
- For example, no "style" for any data in database

Transaction and ACID



- **Atomicity** – ensures that all operations within the work unit are completed successfully. Otherwise, the transaction is aborted at the point of failure and all the previous operations are rolled back to their former state.
- **Consistency** – ensures that the database properly changes states upon a successfully committed transaction.
- **Isolation** – enables transactions to operate independently of and transparent to each other.
- **Durability** – ensures that the result or effect of a committed transaction persists in case of a system failure.

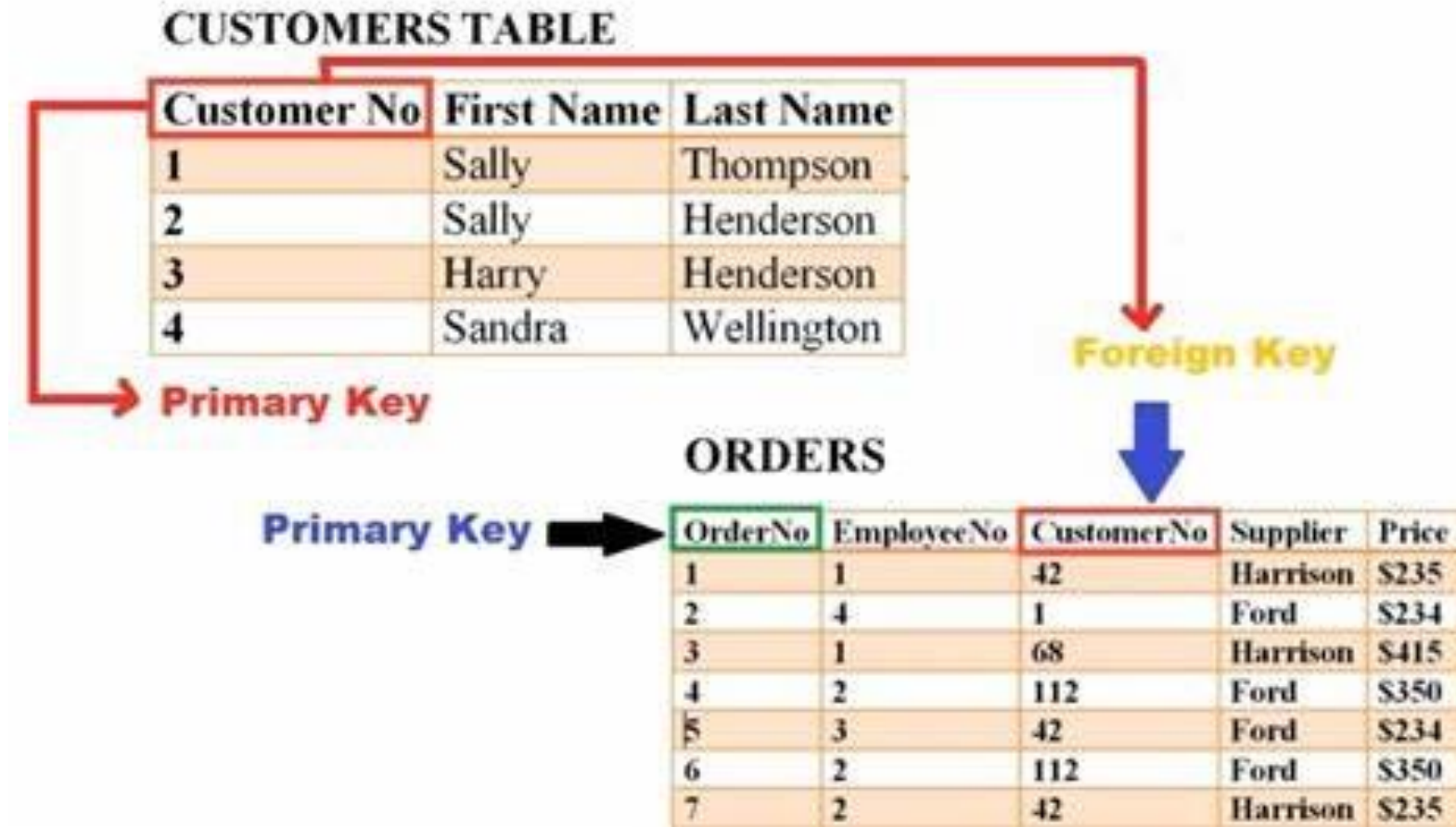
Design database



- The different figures represent different data entities and the specific relationships we have between entities.

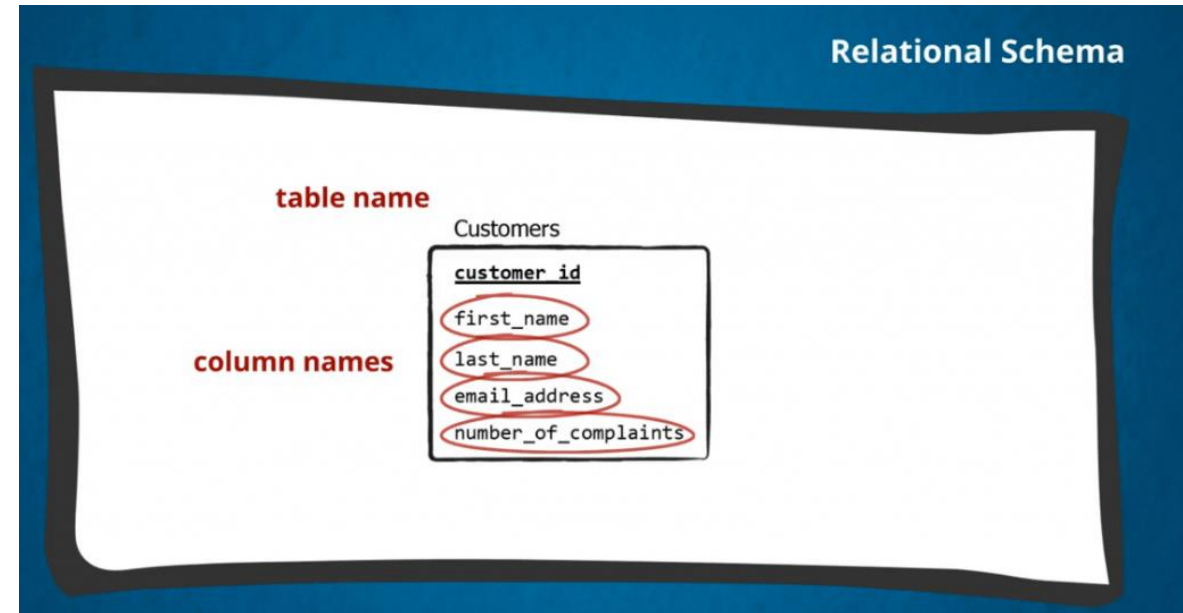
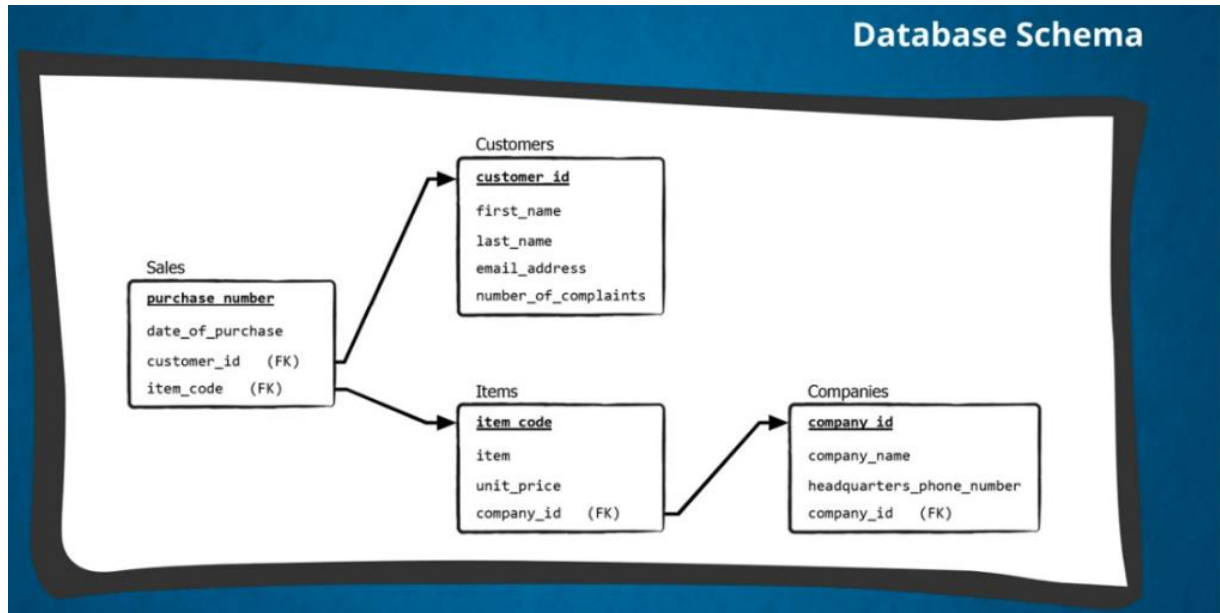
<https://365datascience.com/tutorials/sql-tutorials/sql-why-databases/>

Primary and foreign keys



- **Primary key** – ensures data is unique.
- **Foreign key** – ensures the reference relationship

Design database

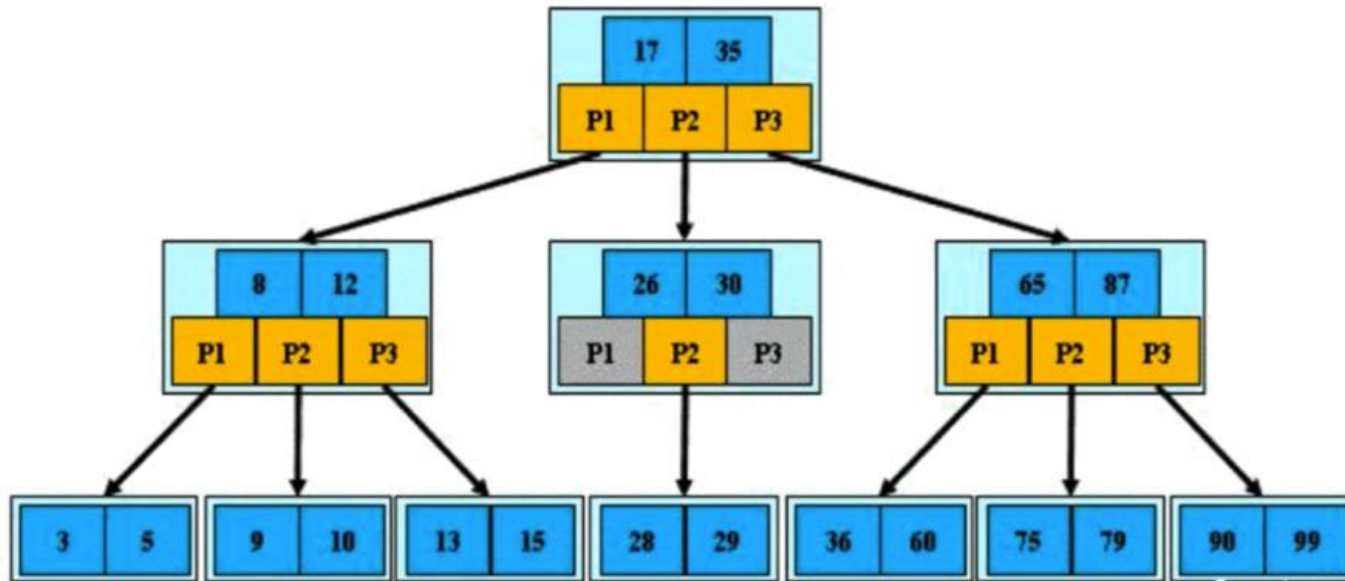


- A table has table name and column names.
- You can also see lines indicating how tables are related in the database.

<https://365datascience.com/tutorials/sql-tutorials/sql-why-databases/>

Index and search

B-Tree

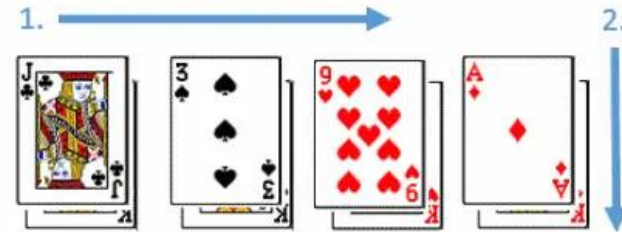


Random Pile of Cards



To Find 8 of Hearts...
Scan 52 Cards (26)
Average Flips = 26

Random Pile of Cards By Suit



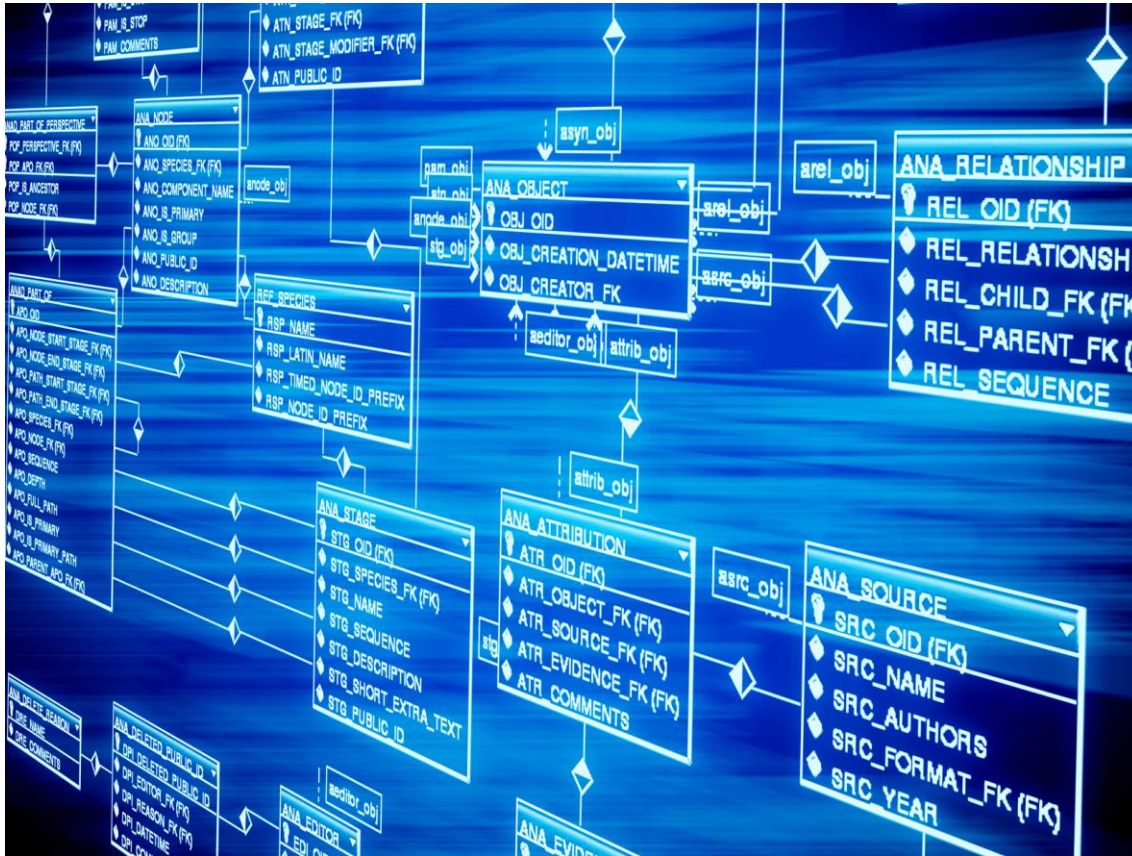
To Find 8 of Hearts...
1. Pick Suit (2)
2. Scan 13 Cards (7)
Average Flips = 9

Creating Piles reduces flip from 26 to 9!

<https://www.zhjynet.cn/archives/2004>

<https://www.essentialsql.com/what-is-a-database-index/>

Operate database - SQL language



Statement 1

Statement 2

Database Explorer 3

```
1 SELECT reported.city,  
2     reported.count_accidents::float /  
3     (reported.count_accidents  
4     + not_reported.count_accidents)::float AS ratio  
5 FROM  
6     (SELECT city, COUNT(*) AS count_accidents  
7     FROM accidents  
8     WHERE reported = 'true'  
9     GROUP BY city) as reported  
10 JOIN  
11    (SELECT city, COUNT(*) AS count_accidents  
12    FROM accidents  
13    WHERE reported = 'false'  
14    GROUP BY city) as not_reported  
15 ON reported.city = not_reported.city  
16 ORDER BY ratio DESC  
17 LIMIT 3;
```

Result 1

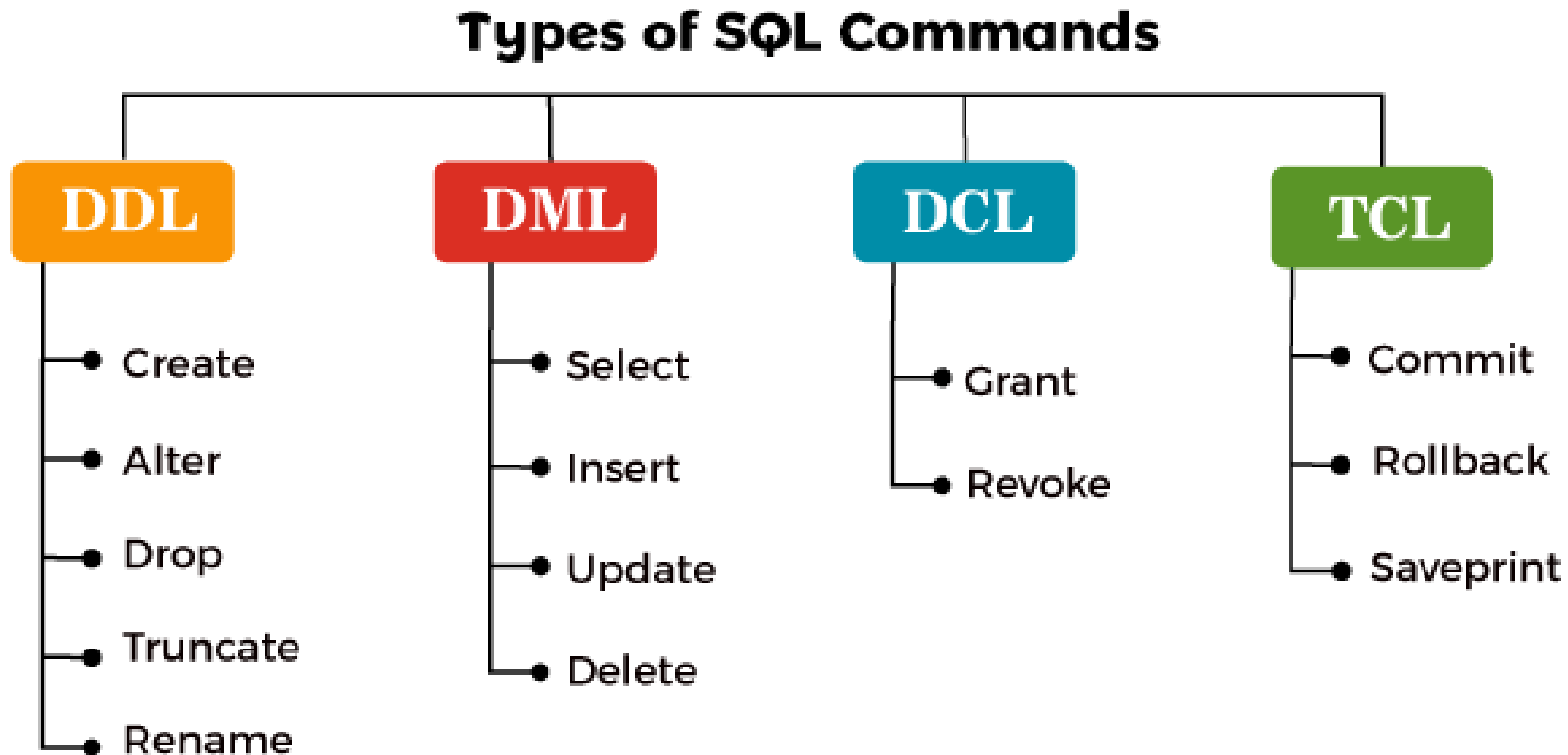
Messages

city	ratio
ROSTOCK	0.67
MAGDEBURG	0.67
BREMEN	0.55

L:16 C:200.21s Timeout:0Max. Rows:01-3/3

- **SQL** or Structured Query Language is a **programming language** used by nearly all [relational databases](#) to query, manipulate, and define data, and to provide access control.

SQL types



- **SQL** or Structured Query Language is a **programming language** used by nearly all [relational databases](#) to query, manipulate, and define data, and to provide access control.

<https://www.javatpoint.com/types-of-sql-commands>

SQL types - DDL

DDL

- Create

- Alter

- Drop

- Truncate

- Rename

```
MySQL 5.5 Command Line Client
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 5.5.62 MySQL Community Server (GPL)

Copyright (c) 2000, 2018, Oracle and/or its affiliates. All rights reserved.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE SCHOOL;
Query OK, 1 row affected (0.01 sec)

mysql> CREATE TABLE t_school(ID INT PRIMARY KEY, School_Name VARCHAR(40), Number_Of_Students INT, Number_Of_Teachers INT,
Number_Of_Classrooms INT, EmailID VARCHAR(40));
Query OK, 0 rows affected (0.14 sec)
```

- DDL stands for data definition language. DDL Commands deal with the schema, i.e., the table in which our data is stored.
- All the structural changes such as creation, deletion and alteration on the table can be carried with the DDL commands in SQL.

SQL types - DML

DML

- Select
- Insert
- Update
- Delete

```
mysql> INSERT INTO t_school(ID, School_Name, Number_of_Students, Number_of_Teachers, Number_of_Classrooms, EmailID) VALUES(1, "Boys Town Public School", 1000, 80, 12, "btps15@gmail.com"), (2, "Guru Govind Singh Public School", 800, 35, 15, "ggps25@gmail.com"), (3, "Delhi Public School", 1200, 30, 10, "dps101@gmail.com"), (4, "Ashoka Universal School", 1110, 40, 40, "aus17@gmail.com"), (5, "Calibers English Medium School", 9000, 31, 50, "cems@gmail.com"), (6, "Cantonment Board High School", 7050, 41, 60, "cbhs@gmail.com"), (7, "Podar International School", 12000, 120, 120, "podaris89@gmail.com"), (8, "Barnes School", 18000, 100, 100, "barnes99@gmail.com"), (9, "D.S Kothari Kanya School", 10000, 120, 125, "ksmh@gmail.com"), (10, "Orchid International School", 20000, 200, 180, "ois@gmail.com");
Query OK, 10 rows affected (0.08 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

- DML stands for Data Manipulation Language. Using DML commands in SQL, we can make changes in the data present in tables.
- Whenever we wish to manipulate the data or fetch the data present in SQL tables, we can use DML commands in SQL.

SQL types - DML

DML

- Select
- Insert
- Update
- Delete

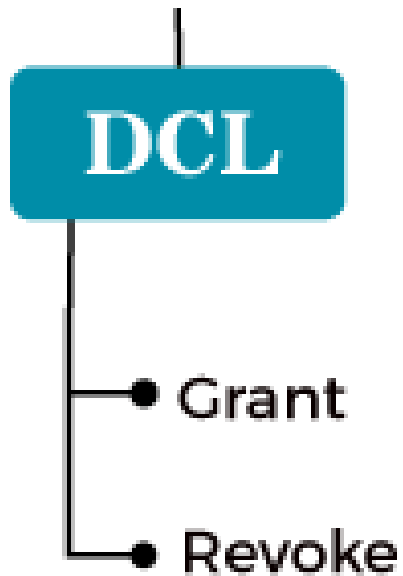
```
SELECT *FROM table_name WHERE condition;
```

```
UPDATE table_name SET column_name = value WHERE condition;
```

```
mysql> DELETE FROM t_school WHERE ID = 6;
```

- It is important to set the **CORRET** where condition

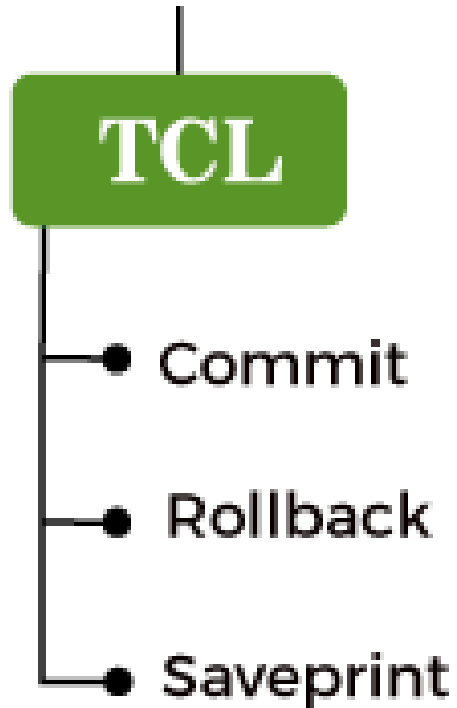
SQL types - DCL



```
mysql> SELECT user, host, account_locked, password_expired FROM user;
+-----+-----+-----+-----+
| user          | host       | account_locked | password_expired |
+-----+-----+-----+-----+
| adam          | localhost  | N              | N                |
| mysql.infoschema | localhost  | Y              | N                |
| mysql.session  | localhost  | Y              | N                |
| mysql.sys      | localhost  | Y              | N                |
| root          | localhost  | N              | N                |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

- Many DBMS like MySQL is a stand alone system with its own users.
- DCL stands for Data Control Language.
- Whenever we want to control the access to the data present in SQL tables, we will use DCL commands in SQL.
- Only the authorized **database users** can access the data stored in the tables.

SQL types - TCL



```
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> INSERT INTO t_school(ID, School_Name, Number_of_Students, Number_of_Teachers, Number_of_Classrooms, EmailID) VALUES(1, "Boys Town Public School", 1000, 80, 12, "btps15@gmail.com"), (2, "Guru Govind Singh Public School", 800, 35, 15, "ggps25@gmail.com"), (3, "Delhi Public School", 1200, 30, 10, "dps101@gmail.com"), (4, "Ashoka Universal School", 1110, 40, 40, "aus17@gmail.com"), (5, "Calibers English Medium School", 9000, 31, 50, "cems@gmail.com");  
Query OK, 5 rows affected (0.00 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> COMMIT;  
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> SAVEPOINT ins;  
Query OK, 0 rows affected (0.00 sec)
```

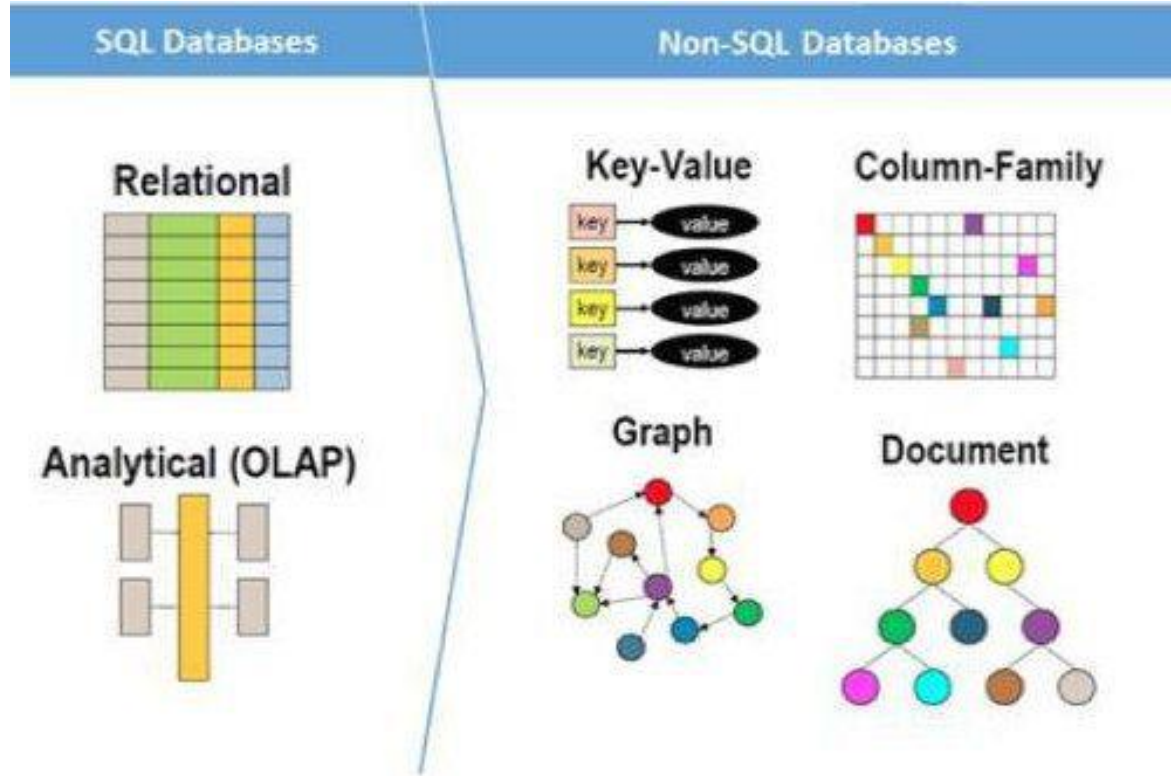
```
mysql> DELETE FROM t_school WHERE ID = 3;  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> ROLLBACK TO ins;  
Query OK, 0 rows affected (0.08 sec)
```

DB will be back to savepoint "ins"

- TCL stands for Transaction Control Language. TCL commands are generally used in transactions.
- Using TCL commands in SQL, we can save our transactions to the database and roll them back to a specific point in our transaction.

Database types

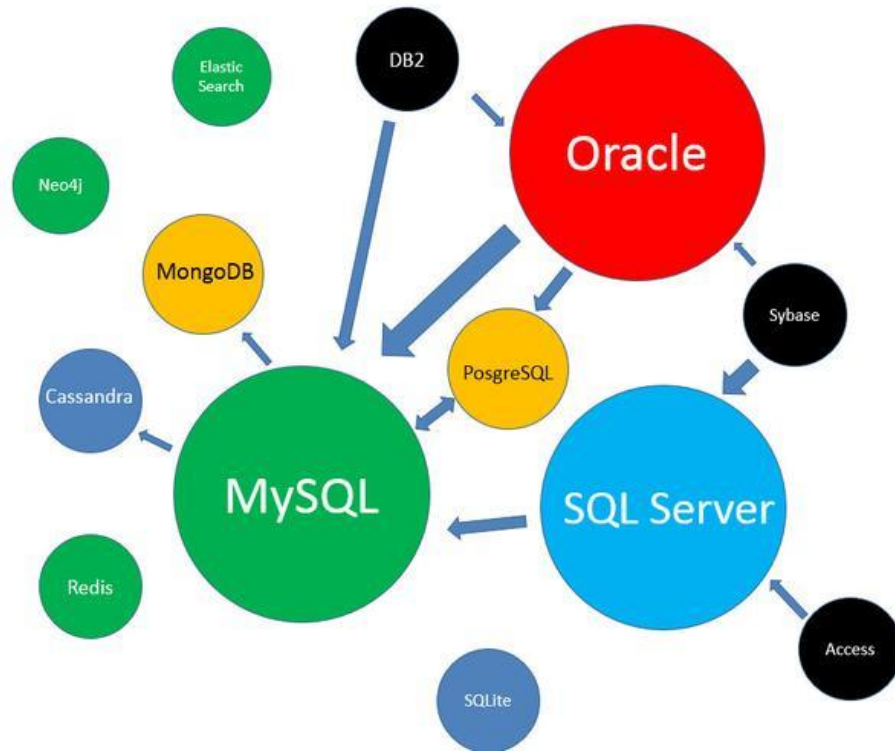


- Be careful, NoSQL means **Not Only** but NOT **no**

<https://www.kdnuggets.com/2016/07/seven-steps-understanding-nosql-databases.html>

<https://www.geeksforgeeks.org/sql-vs-nosql-which-one-is-better-to-use/>

Mainstream DBMS on the market



OceanBase
A Powerful Database System

TDSQL
腾讯分布式数据库

HUAWEI GaussDB
Better DB Better Life

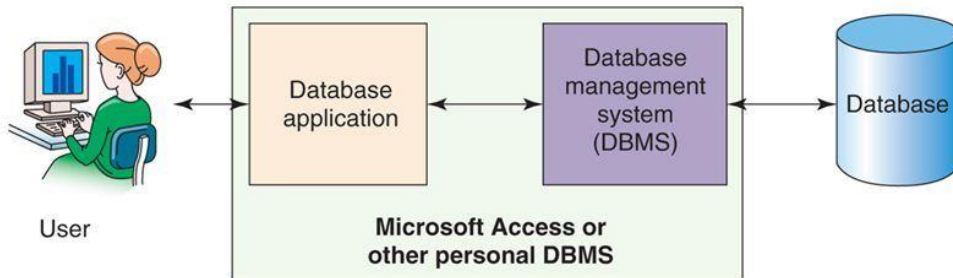
POLARDB

CynosDB

ZTE中兴 GoldenDB 数据库

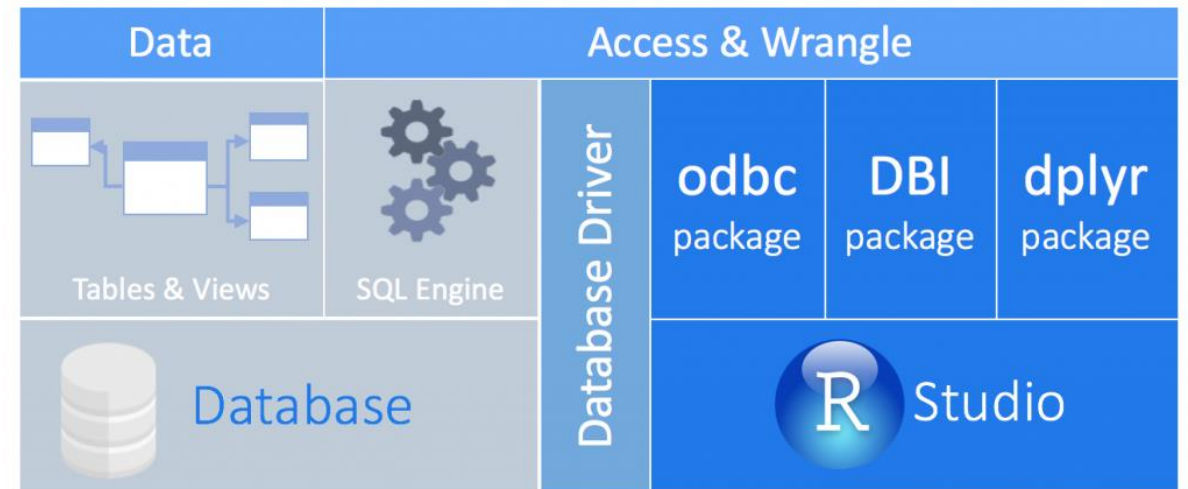
Connect DB

+ Personal Database Systems



1-34

Commercial Databases



<https://dev.mysql.com/doc/refman/5.6/en/transport-protocols.html>

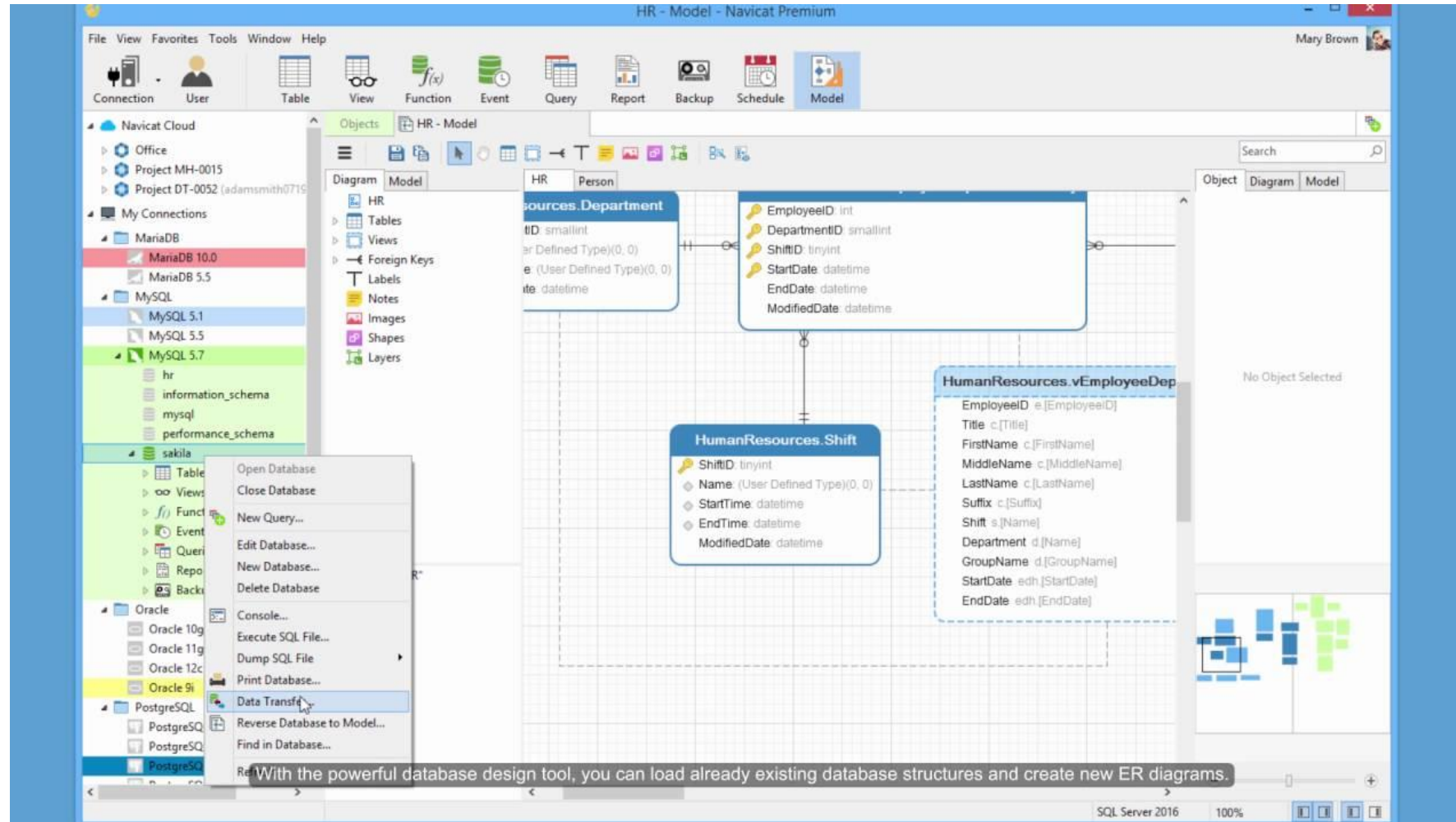
Connect MySQL

<u>--protocol</u> Value	Transport Protocol Used	Applicable Platforms
TCP	TCP/IP	All
SOCKET	Unix socket file	Unix and Unix-like systems
PIPE	Named pipe	Windows
MEMORY	Shared memory	Windows

- **DB is a client/server style software.**
- **DBMS works as the server.**
- **Different DB supports different way to connect, the above shows four transport protocols supported by MySQL.**

<https://dev.mysql.com/doc/refman/5.6/en/transport-protocols.html>

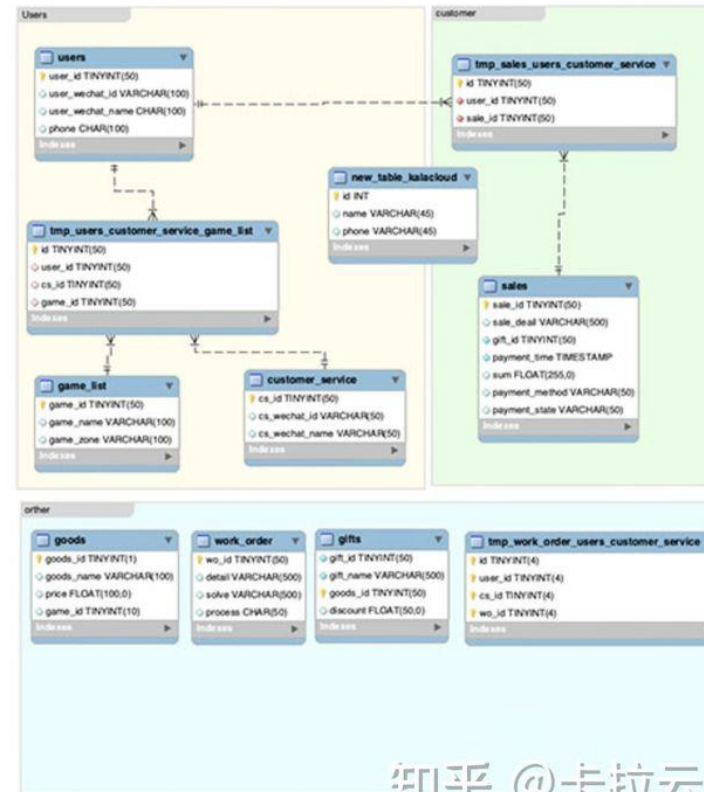
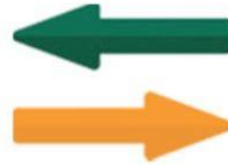
Tools for DB – Navicat



<https://navicat.com/en>

Tools for DB – MySQL Workbench

```
25
26 DROP TABLE IF EXISTS `customer_service`;
27
28 CREATE TABLE `customer_service` (
29   `cs_id` tinyint(50) NOT NULL,
30   `cs_wechat_id` varchar(50) DEFAULT NULL,
31   `cs_wechat_name` varchar(50) DEFAULT NULL,
32   PRIMARY KEY (`cs_id`)
33 ) ENGINE=InnoDB DEFAULT CHARSET=utf8;
34
35 LOCK TABLES `customer_service` WRITE;
36 /*140000 ALTER TABLE `customer_service` DISABLE KEYS */;
37
38 INSERT INTO `customer_service` (`cs_id`, `cs_wechat_id`,
39 VALUES
40 (1,'VIP01','开心小客服'),
41 (2,'VIP02','客服总监'),
42 (3,'VIP_game_super','霸道总裁');
43
44 /*140000 ALTER TABLE `customer_service` ENABLE KEYS */;
45 UNLOCK TABLES;
46
47 kalamcloud.com
48
49 # Dump of table game_list
50 #
51
52 DROP TABLE IF EXISTS `game_list`;
53
54 CREATE TABLE `game_list` (
55   `game_id` tinyint(50) NOT NULL AUTO_INCREMENT,
56   `game_name` varchar(100) DEFAULT NULL,
57   `game_zone` varchar(100) DEFAULT NULL,
58   PRIMARY KEY (`game_id`)
59 ) ENGINE=InnoDB DEFAULT CHARSET=utf8;
60
61 LOCK TABLES `game_list` WRITE;
62 /*140000 ALTER TABLE `game_list` DISABLE KEYS */;
63
64 INSERT INTO `game_list` (`game_id`, `game_name`, `game_zo
65 VALUES
66 (1,'传奇世界','美服'),
67 (2,'梦幻西游','战服'),
68 (3,'征途','全服战争');
69
70 /*140000 ALTER TABLE `game_list` ENABLE KEYS */;
71 UNLOCK TABLES;
72
73
```

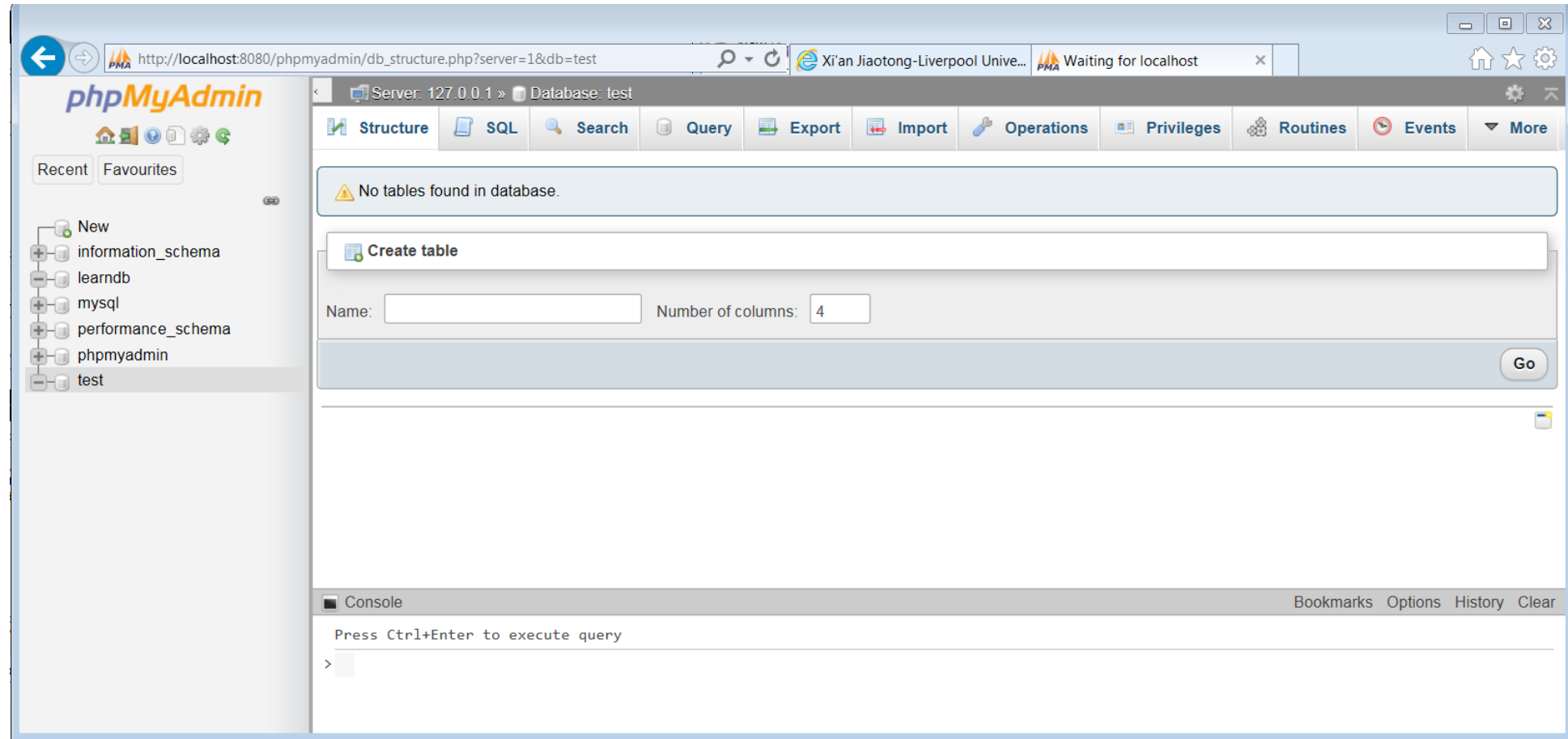


知乎 @卡拉云

<https://www.zhihu.com/question/44003854>

<https://www.mysql.com/products/workbench/>

Tools for DB – PHPMyAdmin (Web Tool)



<http://localhost/phpmyadmin/>

B/S vs. C/S



- C/S means the application would have its own client software.
- For example, you can install "Navicat" to manage your database (through network).
- Web Browsers are also "clients".
- While you can create systems to use the existed web browsers then **NO** need to create your own clients. This architecture is called Browser/Server structure.

Create a table by PHPMyAdmin

Server: localhost » Database: can302 » Table: user

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 first	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 last	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 email	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More

☐ Check all With selected: [Browse](#) [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Spatial](#) [Fulltext](#)

[Add to central columns](#) [Remove from central columns](#)

[Print](#) [Propose table structure](#) [Track table](#) [Move columns](#) [Normalize](#)

[Add](#) 1 column(s) after email [Go](#)

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	id	0	A	No	

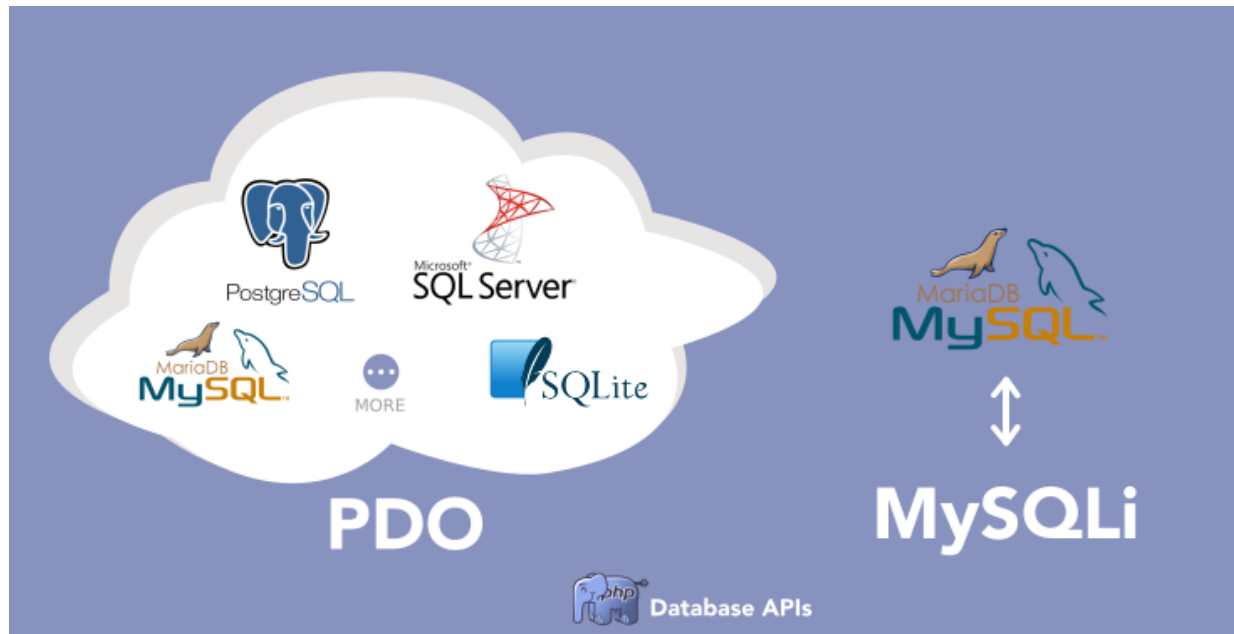
Console Bookmarks Options History Clear

Press Ctrl+Enter to execute query

```
> CREATE TABLE `can302`.`user` ( `id` INT NOT NULL AUTO_INCREMENT , `first` VARCHAR(20) NOT NULL , `last` VARCHAR(20) NOT NUL...
>
```

- Create a database can302
- Create a table user with 4 columns
- Set id as primary and auto-increasement
- All selections translate to SQL command

Manipulate DB by PHP



```
1 // PDO
2 $pdo = new PDO("mysql:host=localhost;dbname=database", 'username', 'password');
3
4 // mysqli, procedural way
5 $mysqli = mysqli_connect('localhost','username','password','database');
6
7 // mysqli, object oriented way
8 $mysqli = new mysqli('localhost','username','password','database');
```

	PDO	MySQLi
Database support	12 different drivers	MySQL only
API	OOP	OOP + procedural
Connection	Easy	Easy
Named parameters	Yes	No
Object mapping	Yes	Yes
Prepared statements	Yes	No
Performance	Fast	Fast
Stored procedures	Yes	Yes

<https://code.tutsplus.com/pdo-vs-mysqli-which-should-you-use--net-24059t>

Manipulate DB by mysqli

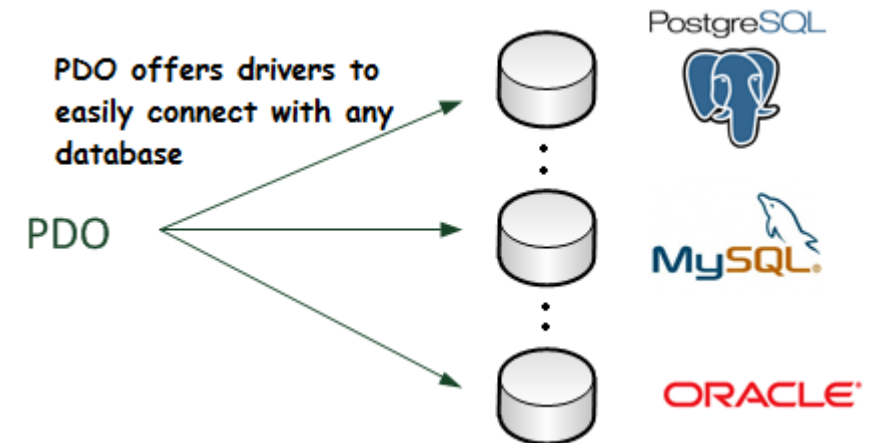
```
1 <?php
2 $servername = "localhost";
3 $username = "username";
4 $password = "password";
5 $dbname = "myDB";
6
7 // Create connection
8 $conn = new mysqli($servername, $username, $password, $dbname);
9 // Check connection
10 if ($conn->connect_error) {
11     die("Connection failed: " . $conn->connect_error);
12 }
13
14
15 $sql = "SELECT id, firstname, lastname FROM MyGuests";
16 $result = $conn->query($sql);
17
18 if ($result->num_rows > 0) {
19     // output data of each row
20     while($row = $result->fetch_assoc()) {
21         echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " . $row["lastname"]. "<br>";
22     }
23 } else {
24     echo "0 results";
25 }
26
27 $conn->close();
28
29 ?>
30
```

- mysqli is a PHP extension to operate MySQL database ONLY.
- It supports both procedure and object oriented paradigm.

Manipulate DB by PDO

```
1 <?php
2
3 //open database by PDO
4 $dbms='sqlite'; //DBMS type
5 $host=''; //Host name
6 $dbName='mysqlitedb.db'; //database name
7 $user=''; //database user
8 $pass=''; //database password
9 $dsn="$dbms:$dbName";
10
11 try {
12     $con = new PDO($dsn, $user, $pass);
13 } catch (PDOException $e) {
14     die ("Error!: " . $e->getMessage() . "<br/>"); //stop the rest code
15 }
16
17 echo '<ul class="list-unstyled components">';
18 $sql = "SELECT * FROM product_category ORDER BY priority";
19 $query = $db_con->query($sql);
20 foreach($query as $row){
21     echo '<li>';
22     echo '<a href="?category='.$row["id"].'">'.$row["name"].'</a>';
23     echo '</li>';
24 }
25 echo '</ul>';
26
27 ?>
28
```

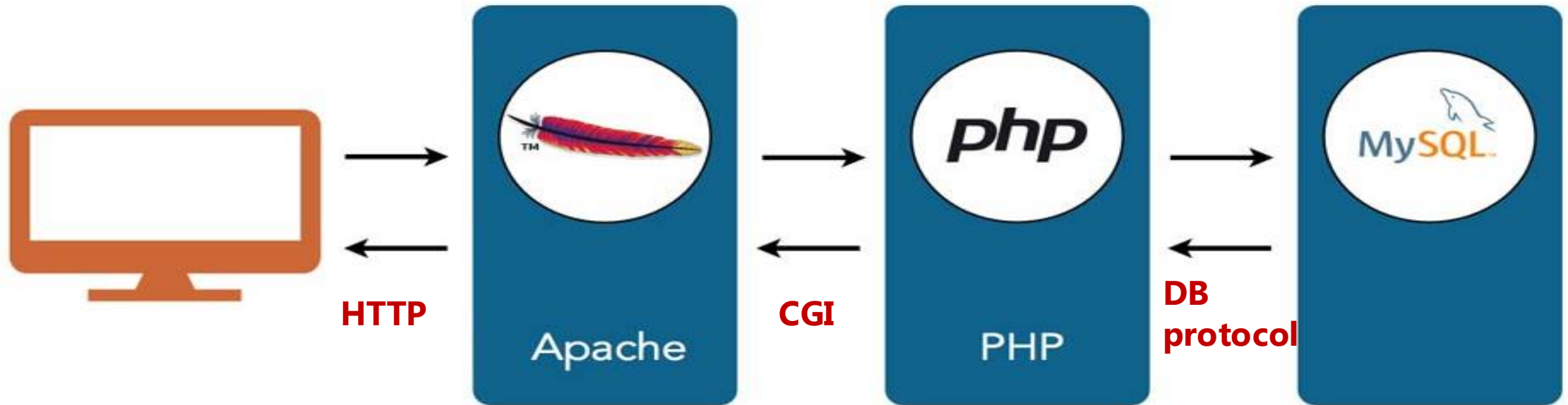
- PDO is a PHP extension to operate many different type database.
- It ONLY supports object oriented paradigm.



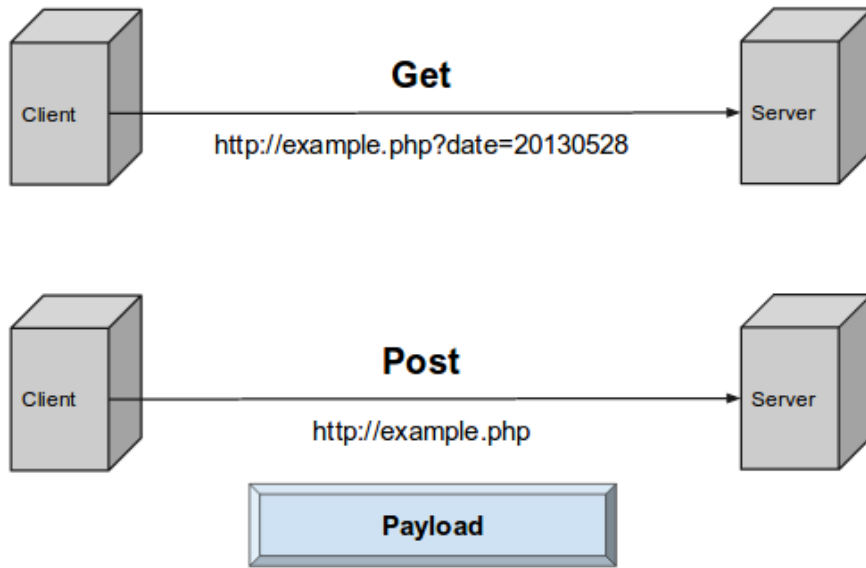
<https://zhuanlan.zhihu.com/p/601407287>

<https://www.php.net/manual/en/book.pdo.php>

Let's wrap them up



HTTP – GET POST



Form Data, JSON Strings, Query Parameters, View States, etc

GET

In GET method, values are visible in the URL.

GET has a limitation on the length of the values, generally 255 characters.

GET performs are better compared to POST because of the simple nature of appending the values in the URL.

This method supports only string data types.

GET results can be bookmarked.

GET request is often cacheable.

GET Parameters remain in web browser history.

POST

In POST method, values are not visible in the URL.

POST has no limitation on the length of the values since they are submitted via the body of HTTP.

It has lower performance as compared to GET method because of time spent in including POST values in the HTTP body.

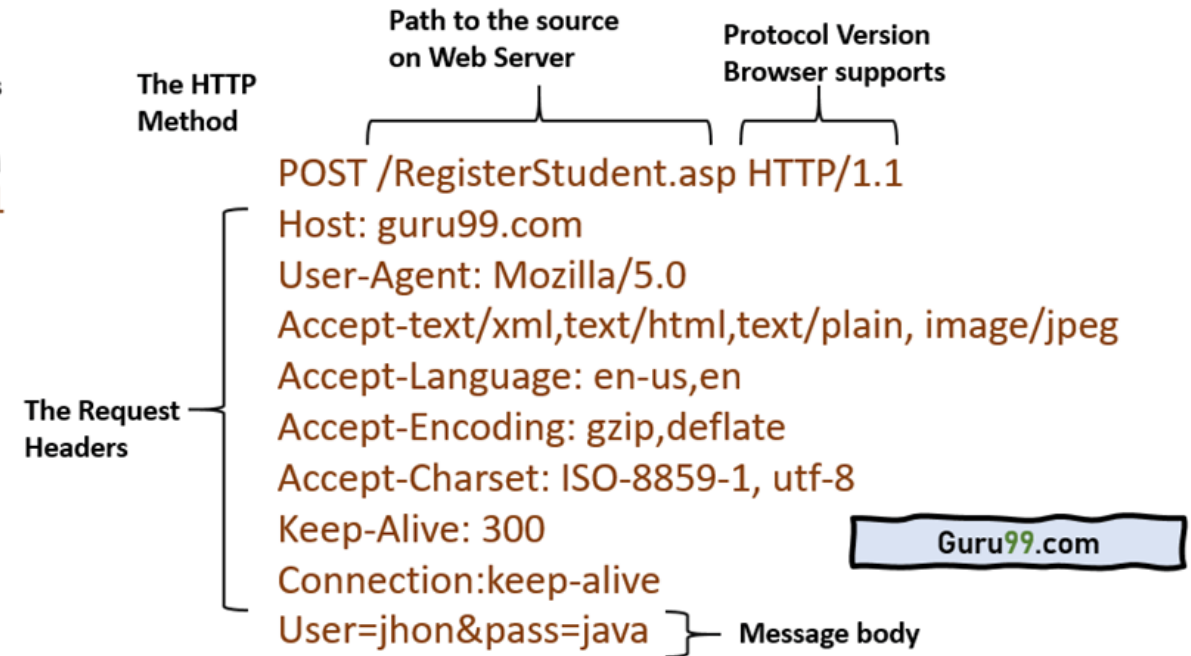
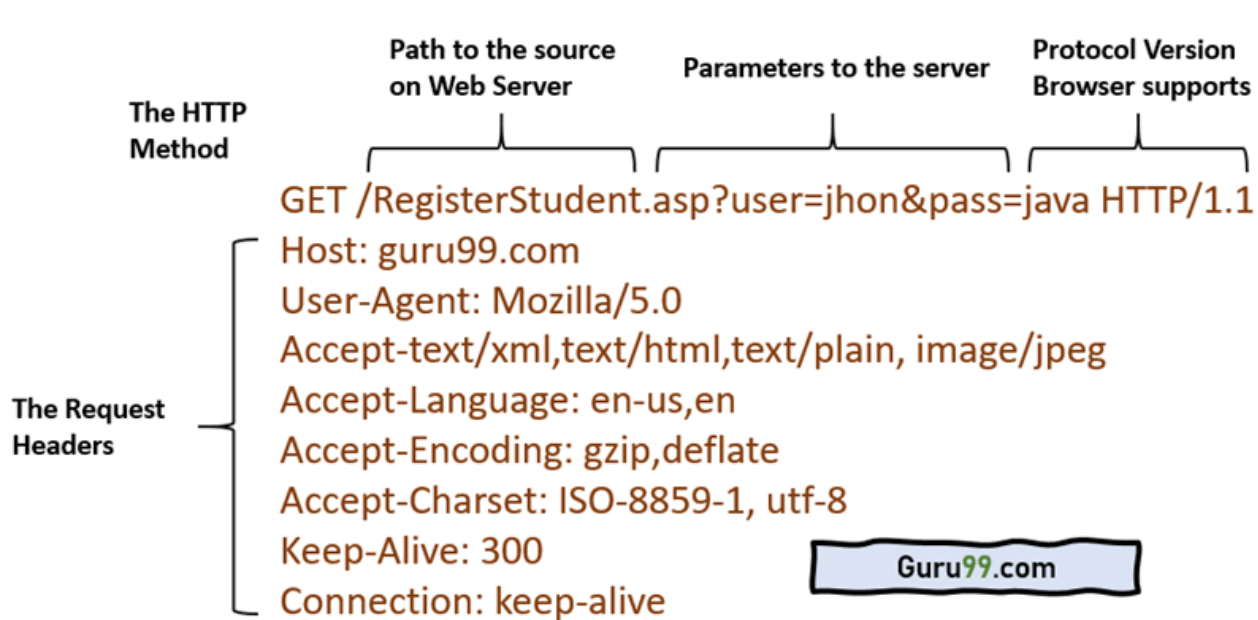
This method supports different data types, such as string, numeric, binary, etc.

POST results cannot be bookmarked.

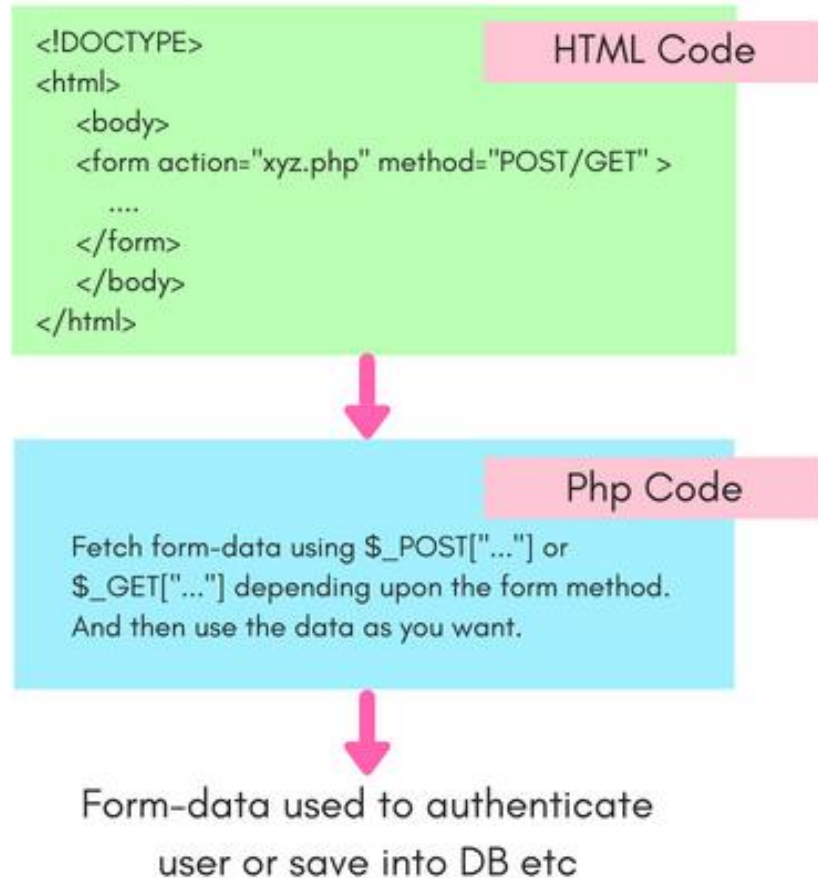
The POST request is hardly cacheable.

Parameters are not saved in web browser history.

HTTP – GET POST

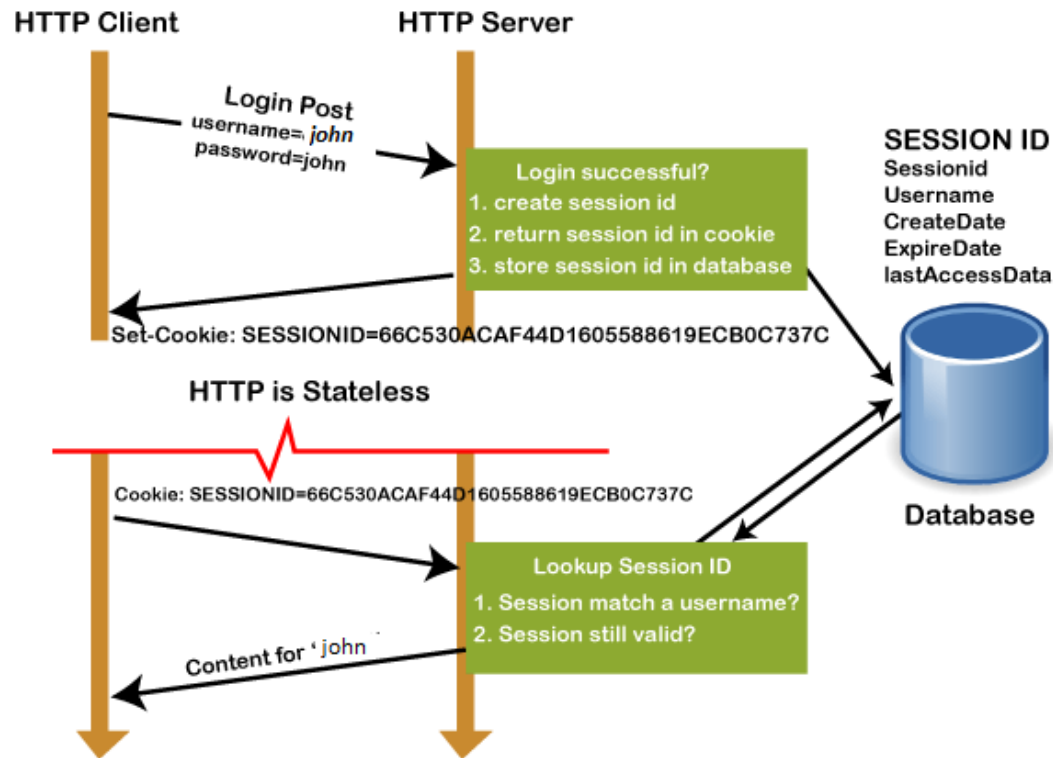


PHP – GET POST



- You can generate the URL with GET parameter directly
- `<form>` can generate both GET and POST request
- PHP uses `$_GET[]` and `$_POST[]` to fetch the HTTP request parameters
- Can one http request contain both GET and POST parameters?

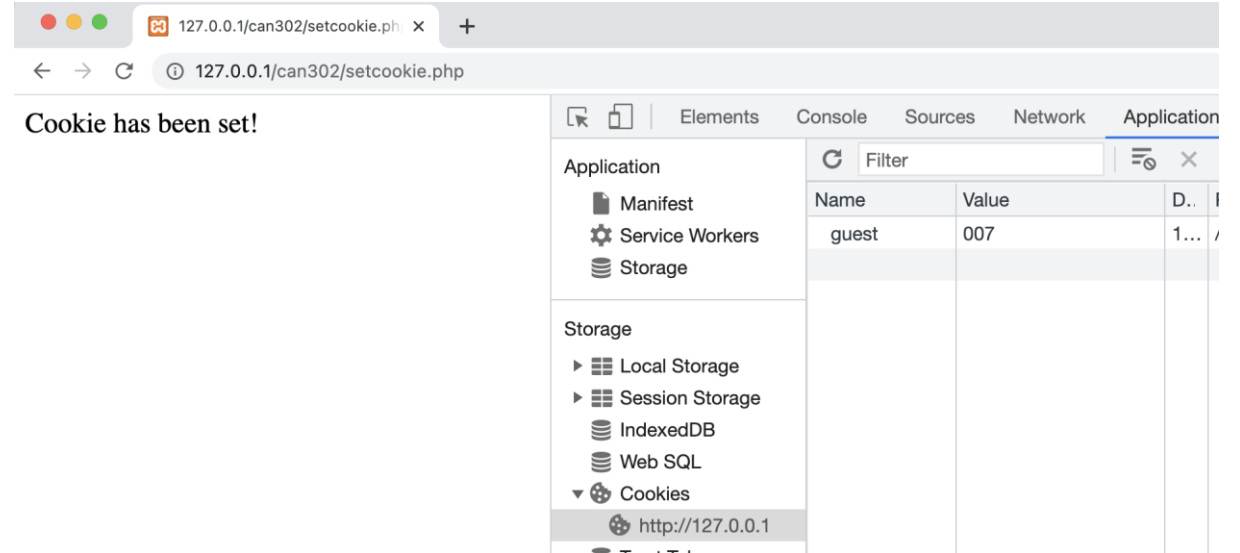
HTTP – Cookie Session



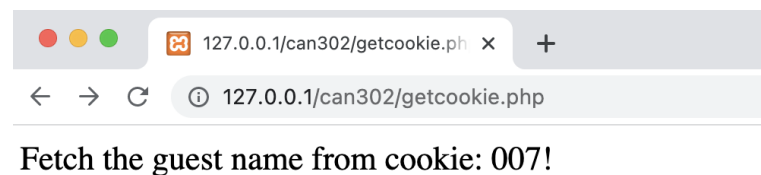
- HTTP is designed to be stateless to be simple, scalable and reliable.
- Stateless meaning that the server does not keep any data (state) between two requests.
- We need to use Cookie and Session to track the status.
- A cookie is a small text file that is stored on the user's computer (Due to the security reason, the browsers only allow interaction with cookie but not other files).
- Cookies are created and shared between the server and browser with the help of an HTTP header.
- Session is a global variable stored on the server. Each session is assigned a unique id which is used to retrieve stored values and the session id is stored as a **cookie** on the user's computer.
- But the information stored within cookies is **NOT** secure because this information is stored in text-format on the client-side, which can be read even tampered by malicious code.

Cookie by PHP

```
1 <?php
2
3 $expire = time() + 60*60*24;
4 setcookie("guest", "007", $expire);
5 echo "Cookie has been set!";
6
7 ?>
```



```
1 <?php
2
3 if (isset($_COOKIE["guest"]))
4     echo "Fetch the guest name from cookie: " . $_COOKIE["guest"] . "<br>";
5 else
6     echo "No cookie!";
7
8 ?>
```



Try them by yourself!

Session by PHP

```
23 session_start(); // magic word to start/reuse a session.
24
25 if (isset($_POST['login'])) {
26     $username = mypost('username');
27     $password = mypost('password');
28     $sql = "SELECT count(*) FROM user WHERE username='$username' AND password='$password'";
29     $count = $con->query($sql)->fetchColumn();
30     if ($count==1) {
31         $_SESSION['username'] = $username;
32     } else {
33         echo '<script type="text/javascript"> alert("Username or password error!"); </script>';
34     }
35 }
36
37 function login(){
38     if(!empty($_SESSION['username'])) {
39         echo "<h2> Welcome " . $_SESSION['username'] . " <a href='logout.php'> logout </a> </h2>";
40     } else {
41         echo <<<EOT
42         <h2> </h2>
43         <div class="col-lg-8" style="text-align:center"><!-- container begin -->
44             <form action="" class="well" method="post"><!-- form begin -->
45                 <h2 class="form-login-heading text-center"> Login demo </h2>
46                 <br>
47                 <input type="text" class="form-control" placeholder="Username" name="username" required>
48                 <br>
49                 <input type="password" class="form-control" placeholder="Your Password" name="password" required>
50                 <br>
51                 <button class="btn btn-primary" type="submit" name="login">Login</button>
52             </form><!-- form finish -->
53         </div><!-- container finish -->
54         EOT;
55     }
56 }
57
58 ?>
59 <!DOCTYPE html>
60 <html lang="en">
61 <head>
62     <meta charset="UTF-8">
63     <meta name="viewport" content="width=device-width, initial-scale=1">
64     <title>CAN302 Login Demo</title>
65     <link rel="stylesheet" href="styles/bootstrap-337.min.css">
66     <script src="js/jquery-331.min.js"></script>
67     <script src="js/bootstrap-337.min.js"></script>
68 </head>
69 <body>
70
71 <?php login() ?>
72
73 </body>
74 </html>
75
```

```
logout.php - CAN302
1 <?php
2
3 session_start();
4 session_destroy();
5
6 $url = $_SERVER['HTTP_REFERER'];
7 echo '<script type="text/javascript"> alert("logout success!"); window.location.href="'.$url.'"; </script>';
8
9 ?>
```

Login demo

'or '1'='1

Login

Try them by yourself!

SQL injection

```
SELECT * FROM
users
WHERE
username = 'Administrator'
AND
password = 'x' OR 'x' = 'x';
```

```
25 if (isset($_POST['login'])) {
26     $username = mypost('username');
27     $password = mypost('password');
28     $sql = "SELECT count(*) FROM user WHERE username= ? AND password= ? ";
29     // $count = $con->query($sql)->fetchColumn();
30     $query = $con->prepare($sql);
31     $query->execute(array($username, $password));
32     $count = $query->fetchColumn();
33     if ($count==1) {
34         $_SESSION['username'] = $username;
35     } else {
36         echo '<script type="text/javascript"> alert("Username or password error!"); </script>';
37     }
38 }
39
```

- **SQL Injection** is a security flaw in **web applications** where attackers insert harmful **SQL** code through user inputs. This can allow them to access sensitive data, change database contents or even take control of the system.
- Using PDO with proper “prepare”, SQL injection can be avoided.

<https://www.geeksforgeeks.org/sql-injection/>

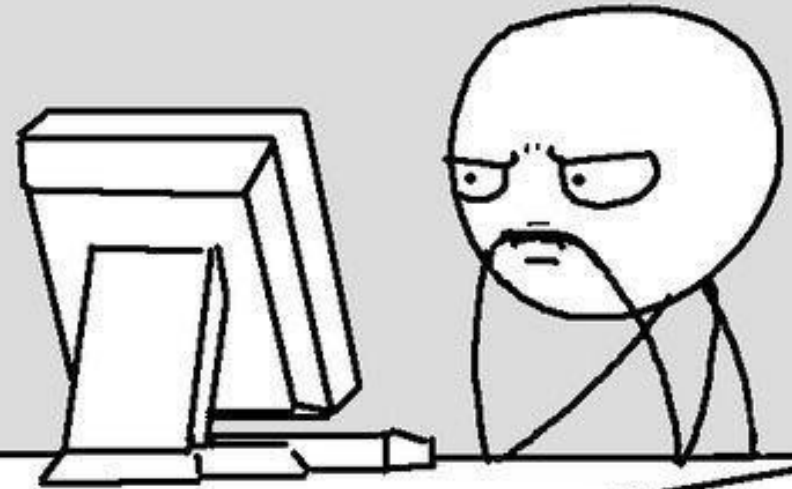
<https://www.php.net/manual/en/pdo.prepare.php>

<https://code.tutsplus.com/pdo-vs-mysqli-which-should-you-use--net-24059t>

Summary

- An introduction to PHP
- An introduction to SQL

It doesn't work..... why?



It works..... why?

